Lecture 9. Decision trees and random forests

Deep Learning for Scientific Discovery

2026/2027

Study route

We assume familiarity with averages, fractions and a table of labelled observations. The probability arguments use independent draws and weighted averages, which are explained where they enter. Derivatives, vectors and covariance provide deeper explanations and can be studied after the numerical examples.
After Sections 1–4, calculate the score of a question, choose a threshold and trace a new observation to its leaf. Section 5 extends the same reasoning to numerical targets. Sections 6–8 explain validation, interpretation and averaging. Section 9 compares all three methods on classification, images and numerical regression. The appendices develop the measurements, proofs and further applications.
Start with `notebook_2026-11-10-00-tree_foundations.ipynb`. Its small tables need no image download. The concentration-response notebook gives a short regression application, and MNIST introduces image classification. Continue with the diabetes regression notebook and the galaxy forests notebook. The galaxy classification notebook also reconstructs the measurements from images and provides the detailed scientific investigation. The README gives the complete route.

Contents

1 Classification by a sequence of questions **3**

2 Galaxy images as a classification problem **6**

2.1 The measurements supplied to the tree 7

3 Decision trees as partitions of measurement space **10**

4 Choosing a split **10**

4.1 Class proportions and Gini impurity 11

4.2 The split criterion 12

4.3 Learning the first galaxy question 13

4.4 Repeating the search inside each child 14

4.5 Prediction for one validation galaxy 15

4.6 Probability predictions and split geometry 17

4.7 Other impurity measures . . . 18

5 Regression trees **18**

5.1 Leaf means and the squared-error criterion . . . 19

5.2 A regression tree for diabetes progression . . . 21

5.3 Error measures . . . 22

6 Tree complexity and validation **23**

6.1 The full descriptor row . . . 23

6.2 Depth selection on the galaxies . . . 25

7 Interpreting a single tree **26**

7.1 Decision paths . . . 26

7.2 Impurity importance . . . 27

7.3 Permutation importance . . . 27

7.4 Measurement artefacts in the descriptors . . . 29

8 Random forests **30**

9 Comparing the methods in applications **30**

9.1 A common fitting and evaluation procedure . . . 31

9.2 Galaxy measurements . . . 31

9.3 Classification directly from images . . . 31

9.4 Numerical regression . . . 32

A Practice feedback **34**

B Image measurements **35**

C Global search and recursive growth **37**

D Misclassification error and entropy **37**

E Further datasets **39**

Symbols and data roles

An observation has measured features $\boldsymbol{x}_i = (x_{i1}, \ldots, x_{id})$ and a target y_i. There are d feature columns and K classes in classification. A node t holds n_t fitting observations. A question compares feature x_j with threshold θ and sends an observation to child L or R. Their fractions are $\pi_L = n_L/n_t$ and $\pi_R = n_R/n_t$. A leaf is a node where the questions end.

Fitting observations determine the questions and leaf values. Validation observations choose settings such as depth. Test observations assess the frozen procedure. A training loss measures fit, while an evaluation score measures the behaviour we care about on new observations. Their numerical values and rankings can differ.

1 Classification by a sequence of questions

Supervised learning builds a predictive rule from labelled examples. The earlier lectures used neural networks, which suit images, sound and other raw signals because they learn their own descriptions of the input from the data. Much of science, however, produces tables, in which each row describes one object and each column holds a quantity measured because it is expected to matter. A greengrocer's record of fruit is such a table, with one row per fruit and columns for its length, its width and the feel of its skin. A catalogue of galaxies with their shapes and sizes and a clinical record of patients with their body mass index and blood pressure have the same form. For tables of this kind a second family of models is at least as widely used as neural networks, the models built from decision trees.

A decision tree predicts the way people sort fruit, by a short sequence of questions, each comparing one measured quantity with a threshold. The questions are learned from labelled examples, and anyone who understands the measurements can read them. A single tree is easy to inspect, but it is unstable, since a few different fruit in the training box can change its questions. Ensemble methods combine many trees into one model. A random forest grows many trees, each on a randomized fitting procedure, and averages their answers, which can make the prediction more accurate and more stable, while the questions of the trees and the importance of each measurement remain open to inspection.

Readable questions matter in science. A model that sorts fruit by elongation and diameter can be checked by anyone who has seen a banana, and a model that sorted them by the colour of the crate they arrived in would be exposed at once. In the same way we want to see which measurements a model of galaxies or patients relies on, check them against what is known, and notice when the model exploits an artefact of the measurement. This chapter develops the tree-based family from a single tree to a forest, with emphasis on what the models learn and how their decisions can be read, and closes with a comparison with a neural network trained on the same data.

Imagine sorting a box of bananas, cherries, pears, peaches and apples. Nobody computes a formula for this. One looks at a fruit and asks a few questions in turn. Is it long and thin? Then it is a banana. If not, is it tiny? Then it is a cherry. If not, does its body narrow into a neck below the stalk? Then it is a pear. Otherwise the skin decides between a fuzzy peach and a smooth apple. Each question looks at one property of the fruit, and its answer decides whether the sorting stops or which question comes next.

A computer needs the properties as numbers, and Figure 1.1 sketches four of them. The elongation x_1 is the length of the fruit divided by its width, close to 1 for round fruit and between 2 and 3 for a banana. The diameter x_2 is the width in centimetres, about 2 for a

cherry and 6 to 8 for the other fruit. The neck taper x_3 compares the width w_{neck} of the fruit just below the stalk with its largest width $w_{\max}$,

$$x_3 = 1 - \frac{w_{\text{neck}}}{w_{\max}}, \tag{1.1}$$

so that it is close to 0 for a round apple and about 0.7 for a pear. The skin flag x_4 equals 1 for a fuzzy skin and 0 for a smooth one. Every fruit is now a row of four numbers $\boldsymbol{x} = (x_1, x_2, x_3, x_4)$, and sorting the fruit means applying a classifier, a rule that turns every such row into the name of a fruit.

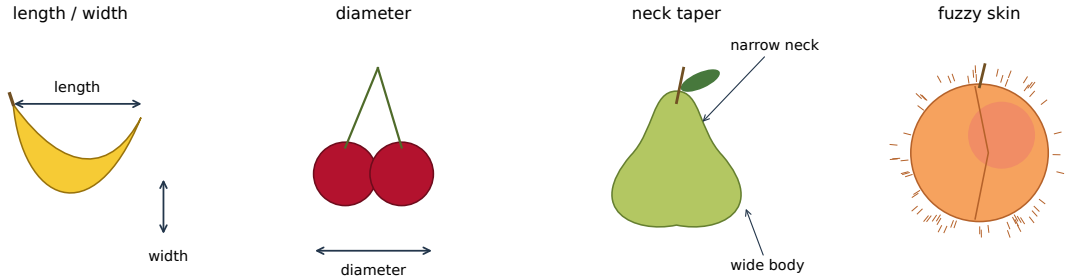


Figure 1.1: The four measurements on which the fruit questions act, the ratio of length to width, the diameter, the neck taper and the fuzziness of the skin.

The greengrocer's questions become comparisons of these numbers with thresholds. A fruit with elongation above 1.8 is a banana. Among the others, a fruit with a diameter below 3 cm is a cherry. Among the remaining ones, a neck taper above 0.5 means a pear, and finally a skin flag of 1 means a peach and 0 an apple. Written as a program, the rule is a nested if-else statement,

```
if length/width > 1.8:   banana
elif diameter < 3 cm:    cherry
elif neck taper > 0.5:   pear
elif fuzzy skin:         peach
else:                    apple
```

and drawn as a graph it is the tree of Figure 1.2. Each internal node asks one question about one measurement, each answer selects the next question, and the last answer reaches a leaf that stores the prediction. Botanists identify plants with keys of this form, and the Galaxy Zoo volunteers who labelled the galaxies of this chapter followed a key written by astronomers.

A fruit is classified by answering the questions from the root down. Figure 1.3 follows a pear with length-to-width ratio 1.4, diameter 7 cm and neck taper 0.7. It answers no to the first question, no to the second and yes to the third, and the third answer ends at the leaf pear. The question about the skin is never asked for this fruit, and its answer could not change the result. A prediction uses only the questions on one path.

Models of this form are called decision trees. A neural network would compute the same kind of rule as a chain of weighted sums of all four numbers at once, while a tree asks about one number at a time. The greengrocer's tree was written by a person who knew in advance which measurements separate the fruit and where the thresholds lie. A decision tree learned from data has to find all of this itself. Training a tree means deciding which question to ask first, about which measurement and at which threshold, which question to ask next in each branch, when to stop asking, and what to predict in each leaf, using only labelled examples. What a trained tree has learned is a set of questions and their order, and it can be read and checked by anyone who understands the measurements.

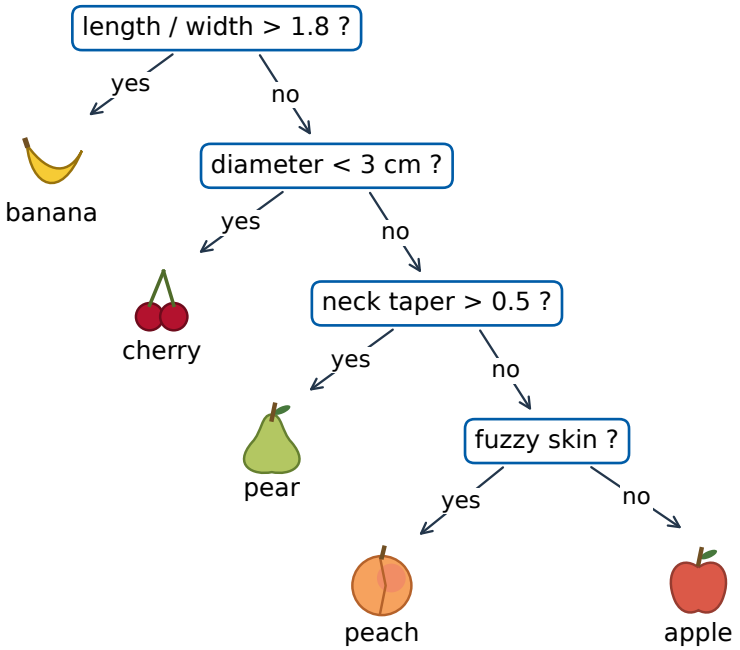


Figure 1.2: A key for sorting five kinds of fruit, written as a decision tree. Every internal node asks one question about one measurement, and every leaf names a class.

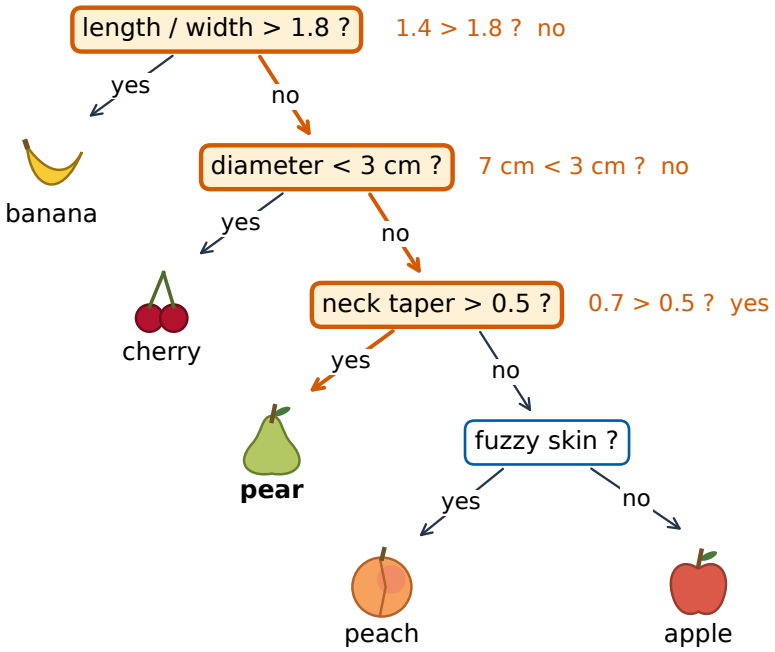


Figure 1.3: The path of one pear through the key. The answers to the three questions it meets are written beside the nodes, and the path ends at the leaf pear.

The rule also has a geometric meaning, shown in Figure 1.4 for a tree of two numerical features. A path from the root to a leaf is a conjunction of conditions, for example $x_1 \le \theta_1$ and $x_2 > \theta_2$, and the inputs that satisfy all conditions of one path form a box with sides parallel to the coordinate axes. The leaves of a tree divide the feature space into disjoint boxes $R_1, \dots, R_L$, and the tree predicts a constant value c_ℓ in each box,

$$f(\boldsymbol{x}) = \sum_{\ell=1}^{L} c_\ell \, \mathbf{1}\{\boldsymbol{x} \in R_\ell\}, \tag{1.2}$$

where $\mathbf{1}\{\cdot\}$ equals one when its condition holds and zero otherwise. Learning a tree means choosing the boxes and the constants.

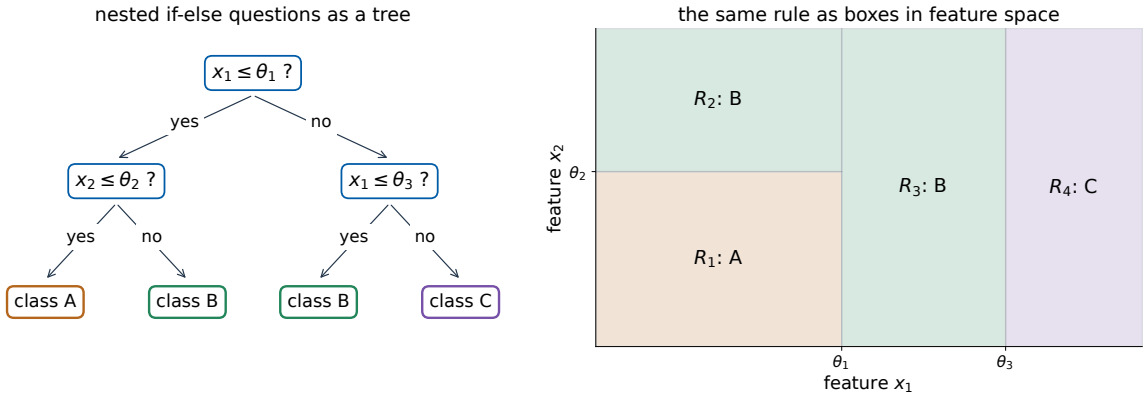


Figure 1.4: A tree of three threshold questions on two features (left) and the four boxes in which it predicts a constant class (right).

A tree does not create its features. It chooses among the features it is given which one to ask about, at which threshold and in which order. The readability of a tree is therefore borrowed from its features. A question about the length-to-width ratio of a fruit means something to every reader, while a question about the brightness of pixel 412 of an image means little, as the pixel trees for handwritten digits in Appendix E show. Building a tree model has two parts. The scientist engineers meaningful features, the ratio and the diameter for fruit, and the algorithm learns the questions. A convolutional network learns its own features from the pixels, which makes it more accurate on images and harder to read.

For fruit the right questions are common knowledge. For galaxy images they are not. Galaxies come as images, so we first turn every image into a few measured descriptors with a physical meaning. Whether the shape of a galaxy is better described by its axis ratio or by its concentration, at which value a round galaxy turns into an intermediate one, and whether a second question helps, then has to be found from labelled galaxies.

2 Galaxy images as a classification problem

Look at the galaxies in Figure 2.2. Some appear nearly round, some moderately elongated, and some narrow and cigar-shaped. We would like to predict one of these three appearance labels from measurements of a new image, which is a classification task. The tree that does this is learned from labelled galaxies [1, 2], and nobody writes a rule such as “very elongated means cigar-shaped” into it.

The images come from Galaxy10 DECaLS, the dataset used in Lecture 6 [3]. The full dataset contains 17,736 colour images of galaxies taken by the DECaLS part of the DESI Legacy

Imaging Surveys [4], in the g, r and z bands. Volunteers of Galaxy Zoo answered a sequence of questions about each galaxy (smooth or featured, edge-on, barred, how tightly wound the spiral arms are), and Galaxy10 turns their votes into ten morphology classes [5]. Figure 2.1 shows one galaxy of each class. We work with the 64-pixel version of Lecture 6, the central 128×128 pixels of each image averaged over 2×2 blocks, which keeps about 0.52 arcseconds per pixel. One image is therefore an array of $64 \times 64 \times 3$ numbers between 0 and 255. The three colour channels are made from the g, r and z exposures and passed through a nonlinear brightness scale chosen for viewing, so a pixel value, which we call a display brightness, rises with the brightness of the sky at that point without being proportional to the light received. Measurements on such images describe shapes well and fluxes only roughly.



Figure 2.1: One galaxy of each of the ten Galaxy10 DECaLS classes. Table 2.1 gives the number of galaxies per class. The three smooth classes used for most of this chapter are coloured.

Table 2.1: Number of galaxies in each Galaxy10 DECaLS class.

Class	Name	Galaxies	Class	Name	Galaxies
0	disturbed	1,081	5	barred spiral	2,043
1	merging	1,853	6	tight spiral	1,829
2	round smooth	2,645	7	loose spiral	2,628
3	intermediate smooth	2,027	8	edge-on without bulge	1,423
4	cigar-shaped smooth	334	9	edge-on with bulge	1,873

The ten classes contain between 334 galaxies (cigar-shaped smooth) and 2,645 galaxies (round smooth). Here we restrict the task to the 5,006 objects labelled round smooth, intermediate smooth, or cigar-shaped smooth. Membership in these three classes is assumed, and a spiral or merging galaxy lies outside the task. The galaxies used to train a model are called fitting galaxies, those used to compare model settings validation galaxies, and those kept for the final scores test galaxies. The present applications focus on the three smooth classes. The labels describe the apparent morphology in the image. The three-dimensional shape of a galaxy and its physical history are separate questions.

2.1 The measurements supplied to the tree

An image is initially an array of pixel values. Before training a tree, we turn it into a short row of numbers. The first feature, the axis-ratio proxy q, describes roundness. Values near one

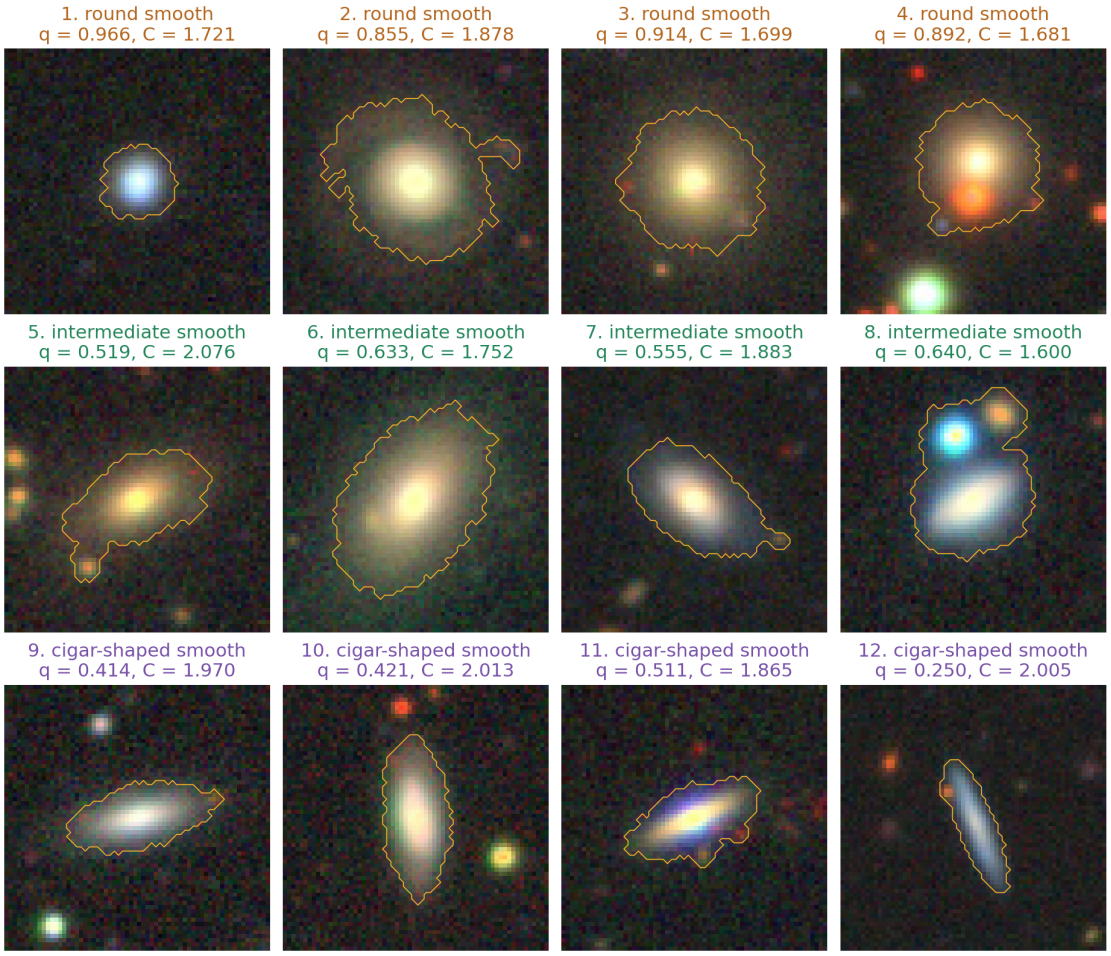


Figure 2.2: Twelve fitting galaxies, numbered to match Table 2.2. They are the first four fitting rows of each selected class, taken before any prediction was made. Orange outlines show the extracted masks. The tree is trained on all 3,407 fitting observations.

indicate a nearly circular brightness distribution, and smaller values indicate elongation. The second, the concentration C, compares the radius enclosing 90% of the measured brightness with the radius enclosing 50%. It describes how that brightness is spread around the centre. Neither feature is a column of Galaxy10. Both are computed from the 64-pixel images. The label comes from the catalogue and is not used in computing either feature.

For now we treat these as supplied columns. Larger q means a rounder image, and larger C means a larger ratio of enclosing radii. Both are dimensionless. We can learn the tree using these columns without knowing how a mask or an eigenvalue is computed. Appendix B derives the measurements and illustrates their limitations. We return to that construction when we investigate the centre offset in Section 7.4.

Table 2.2: The twelve numbered galaxies of Figure 2.2. Row IDs are zero-based positions in the cached dataset. Features are rounded in the table, and fitting uses their full stored precision. All twelve masks succeeded, although success does not guarantee accurate isolation.

Observation	Row ID	Label	q	C
1	3675	round smooth	0.966	1.721
2	4951	round smooth	0.855	1.878
3	4277	round smooth	0.914	1.699
4	3247	round smooth	0.892	1.681
5	7447	intermediate smooth	0.519	2.076
6	6113	intermediate smooth	0.633	1.752
7	6920	intermediate smooth	0.555	1.883
8	6438	intermediate smooth	0.640	1.600
9	7621	cigar-shaped smooth	0.414	1.970
10	7618	cigar-shaped smooth	0.421	2.013
11	7816	cigar-shaped smooth	0.511	1.865
12	7696	cigar-shaped smooth	0.250	2.005

For the full restricted analysis we keep 3,407 fitting observations, 853 validation observations and 746 test observations. The fitting set determines the questions and leaf predictions. Validation selects model settings when alternatives are compared, and test scores describe the resulting models after selection. The test membership is inherited from Lecture 6, and these test galaxies have been seen before in the course, so their scores serve as an illustrative comparison. The galaxies outside the test set were divided into fitting and validation galaxies by stratified sampling, which means that the random division is drawn separately within each class, so that both parts keep the class proportions of the whole. There are 1,797 round, 1,390 intermediate and 220 cigar-shaped objects in the fitting set. In every count vector of this chapter the class order is round, intermediate, cigar-shaped.

One fitting image has a failed mask. It remains in the dataset. Its missing q and C are replaced by the medians of the fitting rows, a replacement of missing values by estimates that is called imputation, and an additional binary failure flag records what happened. The model therefore receives $(q, C, \text{failure flag})$. A minimum of ten observations per leaf prevents the single failed fitting row from becoming a leaf on its own. Subsequent figures show the two shape coordinates with a successful mask. Difficult images stay in the data, and no validation or test value enters the preparation of the fitting data.

3 Decision trees as partitions of measurement space

Before choosing numerical thresholds, we specify the form of the model. A tree is a rooted graph whose positions are called nodes. The root is the first node. An internal node stores a question $x_j \le \theta$ and has two children, the left one for the answer yes and the right one for no. A branch is the edge to a child. A leaf has no children and stores a prediction. The depth of a node is the number of edges from the root, the depth of the tree is the largest leaf depth. Our first galaxy tree is allowed depth two, fixed in advance so that the whole tree can be read, so any prediction involves at most two comparisons. Section 6 chooses the depth from data. The actual feature and threshold at each node still have to be learned.

Training and prediction use different information. During training, the feature rows and known labels tell us whether a question produces useful groups. During prediction, a new object's features answer stored questions, and its unknown label plays no part. Every input follows one path and arrives at one leaf.

The prediction stored at a leaf summarises the fitting observations that fall into its region. Let n_t be the number of fitting observations reaching node t. For a classification task with K classes we count how many of them belong to each class and divide by n_t. The class proportions

$$p_{tk} = \frac{1}{n_t} \sum_{i:\, x_i \in t} \mathbf{1}\{y_i = k\}, \qquad k = 1, \dots, K, \tag{3.1}$$

are nonnegative and sum to one. Here $x_i \in t$ means that observation i reaches node t, and $\mathbf{1}\{\cdot\}$ equals one when its condition holds and zero otherwise. For an input x that reaches leaf ℓ, the tree returns the vector of proportions as its estimate of the class probabilities, and the most frequent class as its hard prediction,

$$\widehat{P}(Y = k \mid x) = p_{\ell k}, \qquad \widehat{y}(x) = \arg\max_k \, p_{\ell k}. \tag{3.2}$$

A leaf containing only one class is called pure. Real image classes overlap, so useful leaves may contain several labels. Their proportions describe the fitting observations in the reached region and serve as probability estimates, whose reliability on new data has to be checked separately. The fitted model consists of the questions, their arrangement and these leaf proportions. A criterion for choosing the questions is still needed.

4 Choosing a split

We start with all 3,407 fitting galaxies at the root. A useful question separates them into children whose label distributions are less mixed, on average, than the parent's. To compare candidate questions we need a measure of mixing and a rule that accounts for the number of observations in each child. The calculation is local to the current node, and no validation or test labels enter it.

We grow the tree one question at a time. At each node we compare the allowed questions on the fitting observations reaching that node and repeat inside its children. This local search is called greedy recursive binary splitting. Appendix C compares it with the global optimization problem.

4.1 Class proportions and Gini impurity

Draw two observations from node t independently and with replacement, and compare their labels. If the node is pure, the two labels always agree. If the classes are mixed, they sometimes disagree, and the disagreement becomes more frequent the more evenly the classes are mixed. The probability of disagreement is the Gini impurity of the node [1],

$$I_G(t) = \sum_{k=1}^{K} p_{tk}(1 - p_{tk}) = 1 - \sum_{k=1}^{K} p_{tk}^2. \tag{4.1}$$

The first form follows by conditioning on the first draw. If it gives class k, which happens with probability p_{tk}, the second draw disagrees with probability $1 - p_{tk}$. The second form counts agreement instead. Both draws give class k with probability p_{tk}^2, and one minus the total agreement probability is the disagreement probability. Figure 4.1 shows the count for three nodes of five galaxies, a pure node of five round galaxies, a node of four round galaxies and one intermediate galaxy, and a node of three round and two intermediate galaxies. Drawing two galaxies from a node of five with replacement gives $5 \times 5 = 25$ equally likely ordered pairs, one for each cell of the grid, with the row giving the first draw and the column the second. The diagonal cells are included because the same galaxy may be drawn twice. A cell is shaded when the two galaxies of its pair carry different labels, and the fraction of shaded cells is I_G. The three nodes have 0, 8 and 12 disagreeing pairs out of 25, and (4.1) gives the same values,

$$1 - 1^2 = 0, \qquad 1 - \left(\tfrac{4}{5}\right)^2 - \left(\tfrac{1}{5}\right)^2 = \tfrac{8}{25} = 0.32, \qquad 1 - \left(\tfrac{3}{5}\right)^2 - \left(\tfrac{2}{5}\right)^2 = \tfrac{12}{25} = 0.48. \tag{4.2}$$

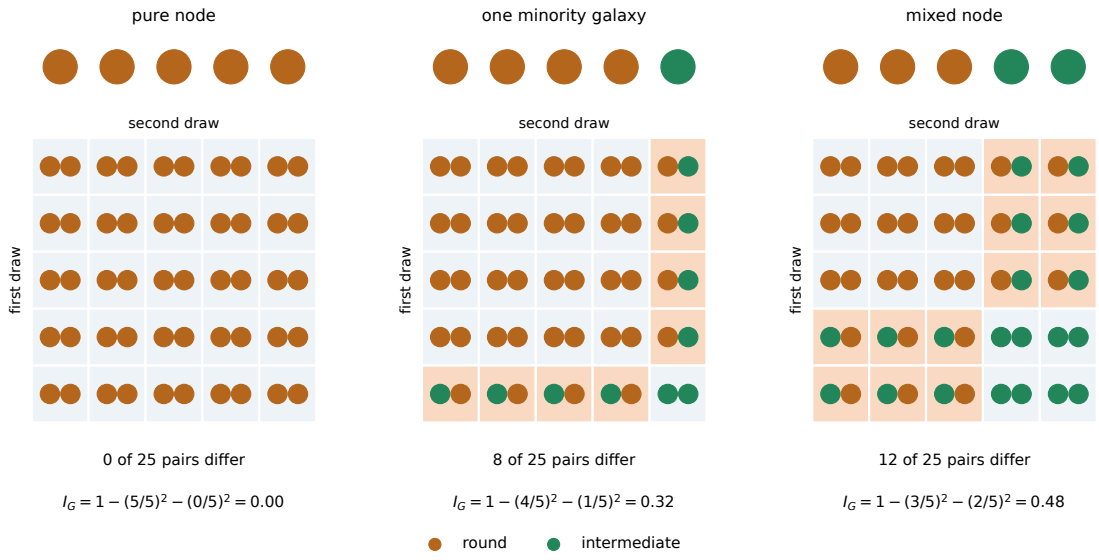


Figure 4.1: Gini impurity as the fraction of disagreeing pairs. Above each grid, a node of five galaxies (brown round, green intermediate). Each cell of the 5×5 grid is one equally likely ordered pair of draws with replacement, first draw in the row and second in the column, shown as two coloured dots. Orange cells mark pairs with different labels, 0, 8 and 12 of 25, which equal $I_G = 0$, 0.32 and 0.48.

The impurity vanishes for a pure node and is largest for equal proportions, where $p_{tk} = 1/K$ and

$$0 \le I_G(t) \le 1 - \frac{1}{K}, \qquad I_G = 2p(1-p) \text{ for two classes with proportions } p,\ 1-p. \tag{4.3}$$

Practice 1. Mixing in a node

A node has four observations of class A and one of class B. Calculate its class probabilities, hard prediction and Gini impurity. Distinguish the probability of error of the majority prediction from the probability that two independent labels disagree. Attempt this before reading the feedback in Appendix A.

4.2 The split criterion

A candidate question $x_j \le \theta$ sends n_L observations of node t to the left child L and n_R to the right child R, with $n_t = n_L + n_R$. We require both children to be nonempty. The children have their own proportions and impurities. A small pure child next to a large mixed one should count for little, since most observations remain in the mixed group, so we weight each child by the fraction of the parent's observations it receives, $\pi_L = n_L/n_t$ and $\pi_R = n_R/n_t$. The weighted child impurity is the expected impurity at the destination of an observation drawn at random from the parent. The decrease in impurity

$$\Delta I(t; j, \theta) = I(t) - \pi_L\, I(L) - \pi_R\, I(R) \tag{4.4}$$

is the score of the question. Training chooses, among the admissible features j and thresholds θ, the pair with the largest decrease. A question is admissible when both of its children satisfy the size rules fixed before training, for instance a minimum of ten observations per child. Figure 4.2 shows why the weights matter, for a node of six round and four intermediate galaxies with $I_G = 0.48$. The first question divides it into two halves, four round and one intermediate on one side and two round and three intermediate on the other, each with weight $\pi = 0.5$ and impurities 0.320 and 0.480. The second question splits off one round galaxy, a pure child with impurity zero but weight $\pi_L = 0.1$, and leaves five round and four intermediate galaxies in a child of weight $\pi_R = 0.9$ whose impurity 0.494 is close to that of the parent. By (4.4),

$$\Delta I_G^{(1)} = 0.48 - 0.5\cdot 0.320 - 0.5\cdot 0.480 = 0.080, \qquad \Delta I_G^{(2)} = 0.48 - 0.1\cdot 0.000 - 0.9\cdot 0.494 = 0.036. \tag{4.5}$$

The pure child of the second question contributes little because it carries a tenth of the galaxies, and the first question, which changes the composition of both large halves, is the better one.

Only finitely many distinct fitting partitions need to be examined. Sort the fitting values of feature j in the node. All thresholds that lie between two neighbouring distinct values produce the same division of the observations, so one representative per gap suffices, and the usual choice is the midpoint. A feature with n_t distinct values in the node offers at most $n_t - 1$ candidate thresholds, and after sorting, the class counts of the two children can be updated as the threshold moves from one gap to the next. With d features, the search over all of them costs of the order of $d\, n_t \log n_t$ operations per node.

Practice 2. Comparing questions

A parent has six A and four B labels. Question U produces counts $(4, 1)$ and $(2, 3)$. Question V produces $(1, 0)$ and $(5, 4)$. Calculate both weighted child impurities and choose a question. State the information needed to evaluate a new candidate threshold. Attempt this before reading the feedback in Appendix A.

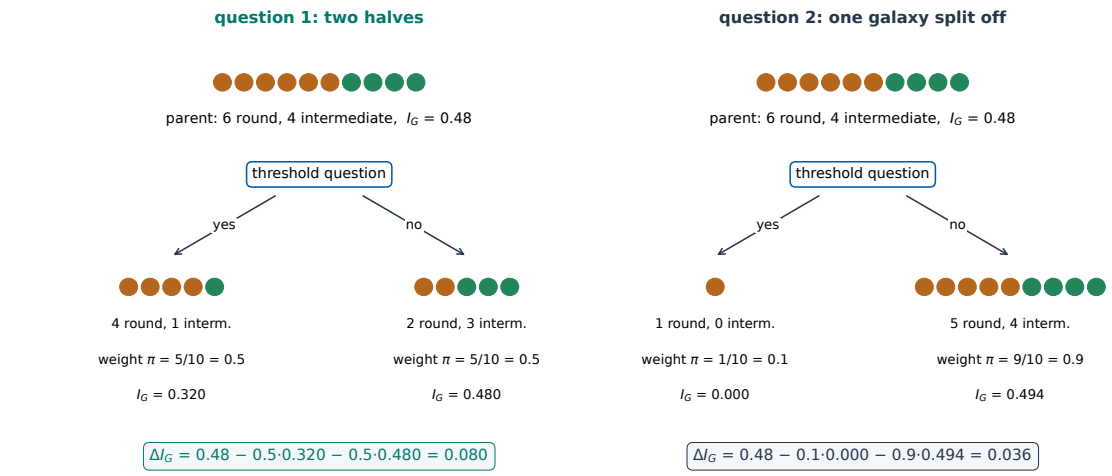


Figure 4.2: Two threshold questions for a node of six round (brown) and four intermediate (green) galaxies, $I_G = 0.48$. Left, the question divides the node into two halves of weight 0.5 with impurities 0.320 and 0.480. Right, the question splits off one round galaxy, a pure child of weight 0.1, and leaves a child of weight 0.9 with impurity 0.494. The weighted decreases are 0.080 and 0.036.

4.3 Learning the first galaxy question

The parent counts are $(1797, 1390, 220)$, so the observed proportions are not equal. Equation (4.1) gives

$$I_G(t) = 1 - \left(\frac{1797}{3407}\right)^2 - \left(\frac{1390}{3407}\right)^2 - \left(\frac{220}{3407}\right)^2 = 0.551183. \tag{4.6}$$

Sort the fitting values of q, try midpoint thresholds between neighbouring distinct values, and count the labels on each side. Repeat for C. Reject a candidate if either child would have fewer than ten observations. The failure flag cannot produce an admissible split because only one fitting row has value one. Figure 4.3 shows the Gini decrease for every admissible threshold on the two shape features. The larger maximum belongs to q, so the first question asks about the axis ratio. The concentration lost this comparison on the fitting galaxies, and it was never excluded in advance.

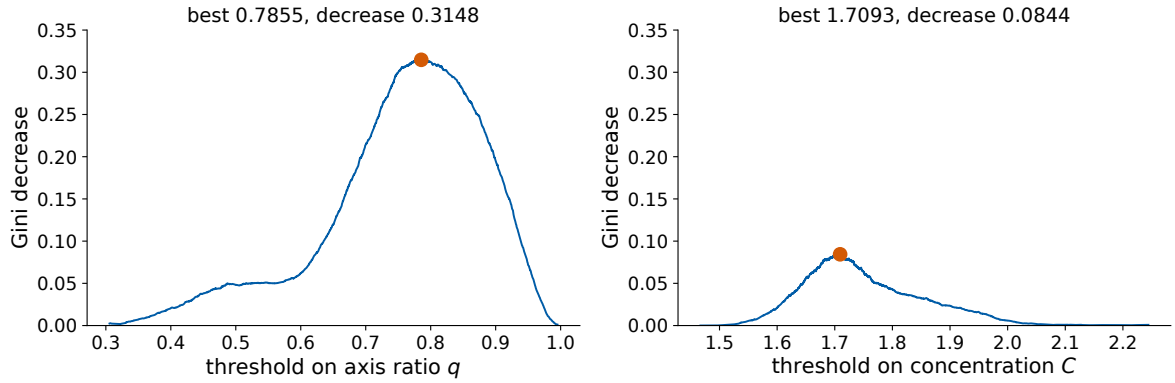


Figure 4.3: Root search on the 3,407 fitting galaxies. Each curve evaluates candidate thresholds while all other choices are held fixed. The best axis-ratio threshold gives a larger decrease than the best concentration threshold. Thresholds are displayed rounded, and the model uses their full precision.

Table 4.1: Best admissible root question for each shape feature. Counts have order round, intermediate, cigar-shaped.

Question	n_L	n_R	Gini decrease
$q \leq 0.785473$	1754	1653	0.314761
$C \leq 1.709295$	1616	1791	0.084387

For the winning question $q \leq 0.785473$, the left child has counts $(203, 1332, 219)$ and the right child $(1594, 58, 1)$. Substituting their proportions into (4.1) gives $I_G(L) = 0.394317$ and $I_G(R) = 0.068880$. The weighted calculation is therefore

$$\Delta I_G = 0.551183 - \frac{1754}{3407}(0.394317) - \frac{1653}{3407}(0.068880) = 0.314761. \tag{4.7}$$

The weighted child impurity is 0.236422. Most round objects have gone right, while most intermediate and cigar-shaped objects remain together on the left. Neither child is pure. A good question need not finish the classification. It makes the next question more useful within each resulting group.

4.4 Repeating the search inside each child

The left child now supplies its own 1,754 rows to the same search. Its best question is $q \leq 0.515985$, which gives counts $(37, 77, 193)$ and $(166, 1255, 26)$. Their majority labels are cigar-shaped and intermediate. The root did not separate these two groups, and a second question within the elongated branch does. The right child uses its own 1,653 rows and chooses $q \leq 0.823698$, giving $(76, 33, 0)$ and $(1518, 25, 1)$. Both of these leaves predict round, but their estimated round-class probabilities differ: $76/109 \simeq 0.697$ and $1518/1544 \simeq 0.983$.

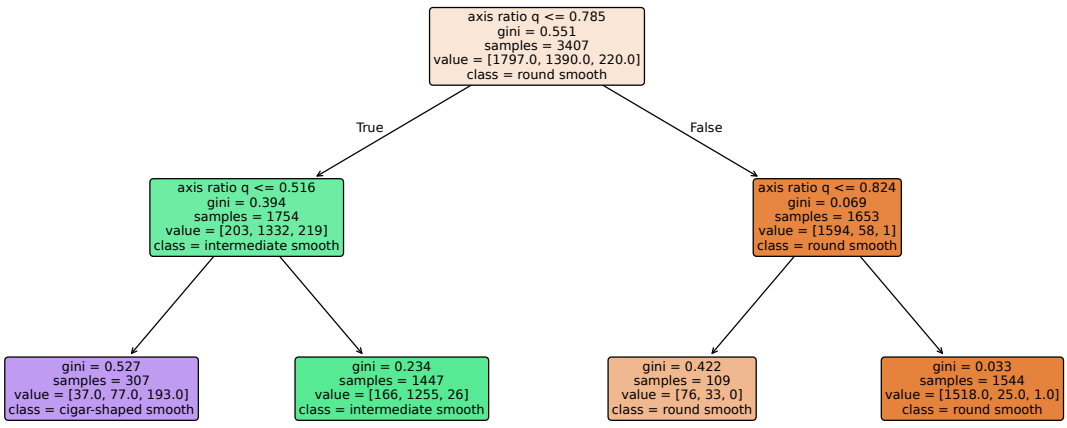


Figure 4.4: The learned depth-two galaxy tree. Each question was chosen from the rows reaching its node. Counts are fitting counts, ordered round, intermediate, cigar-shaped, and "class" is the majority label. Both shape features were eligible, but every chosen question uses q. All four leaves remain mixed and stop at the prescribed depth.

The split on the rounder branch changes the probability estimates without changing either child's majority label. It still decreases Gini impurity because the two label distributions differ. A probability prediction and a hard class decision carry different information, and

Appendix D derives the distinction more generally. A split need not make each child purer. The cigar-shaped leaf has Gini 0.5273, larger than its parent's 0.3943, but the size-weighted impurity of both children decreases. All leaves have reached depth two and each has at least ten fitting rows, so training stops. There was no instruction to make pure leaves, and concentration was not forced into the tree just because it was measured.

Growth stops at a node when the node is pure, when the tree has reached a prescribed maximum depth, or when no question can give both children a prescribed minimum number of observations. These stopping rules, the tie rule and the impurity are choices made before training. The search is greedy. It takes the best question at the current node without looking further ahead, and a sequence of locally best questions need not form the best tree. A known failure is a label that depends on two features jointly, as in the exclusive-or pattern where class A occupies the lower-left and upper-right quadrants. No single threshold on either feature changes the class proportions, every first question has zero decrease, and the useful structure becomes visible only after a first split that appears worthless.

4.5 Prediction for one validation galaxy

Take the first validation galaxy, row 4234, whose measured axis ratio is $q = 0.908681$. It answers no to $q \le 0.785473$ and no to $q \le 0.823698$. The reached leaf contains 1,544 fitting galaxies with counts $(1518, 25, 1)$, so

$$\widehat{\boldsymbol{p}}(x) = \frac{(1518, 25, 1)}{1544} \simeq (0.98316, 0.01619, 0.00065). \tag{4.8}$$

The largest component gives the prediction round smooth. The validation label was not needed to compute this result. The measured feature, the two answers, the reached leaf and its fitting counts form a complete account of how this model produced its prediction.

A path also defines a region of measurement space. A comparison $x_j \le \theta$ selects one side of the boundary $x_j = \theta$, a half-space. If a path to leaf ℓ contains s comparisons and $\mathcal{H}_r$ is the half-space selected by answer r, then

$$R_\ell = \bigcap_{r=1}^{s} \mathcal{H}_r. \tag{4.9}$$

For the leaf just reached, $\{q > 0.785473\} \cap \{q > 0.823698\} = \{q > 0.823698\}$, with no restriction on C. Figure 4.5 shows the four resulting strips in the (q, C) plane. All inputs within one strip receive the same probability vector. In general, coordinate-threshold questions create axis-aligned rectangles or higher-dimensional boxes, possibly unbounded. An oblique boundary such as $q + C = 2$ requires a staircase of such regions. The leaves partition the input space into disjoint regions whose union covers it.

Boxes with sides parallel to the axes suit some problems and not others, and Figure 4.6, modelled on Figure 8.7 of [2], shows the difference on two synthetic tasks. Each task has 500 training points and 5,000 test points spread uniformly over a square, with two classes. In the first, the classes are separated by the oblique line $x_1 + x_2 = 0$. In the second, one class fills the box $x_1 > 0.2$, $x_2 < 0.1$. Logistic regression, a single linear unit with a sigmoid output as in the one-layer networks of the early lectures, draws one straight boundary, and a tree of depth four draws up to sixteen boxes. On the oblique task the linear model reaches a test accuracy of 0.974 and the tree 0.935, since the tree must imitate the line by a staircase. On the box task the tree reaches 0.999 and the linear model 0.928, since no straight line follows the corner of the box. Which model suits a problem depends on the shape of the boundary

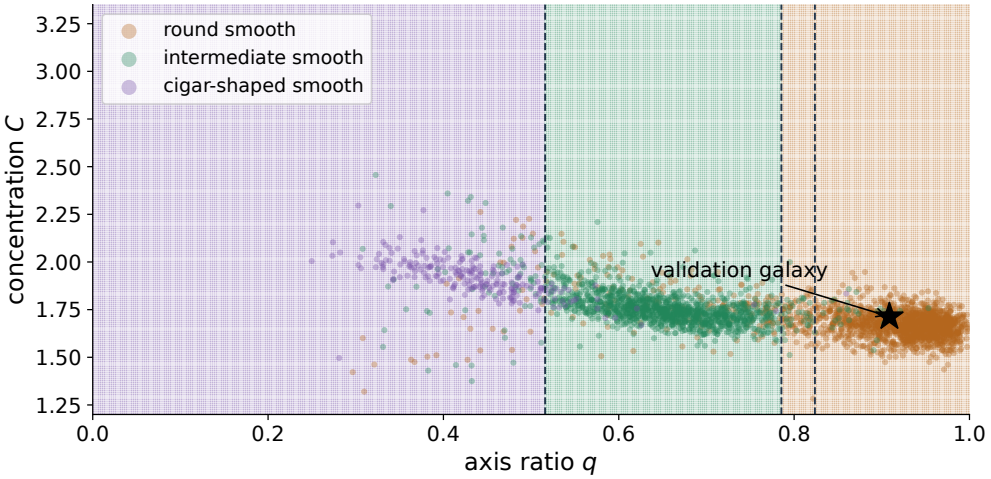


Figure 4.5: The fitting galaxies in shape-feature space, with the learned boundaries. Concentration varies within every leaf, but this depth-two tree does not use it. Two neighbouring right-hand regions share a majority label while retaining different class proportions. The displayed boundaries describe the fitted model and mark no physical division between galaxy populations.

between the classes. A network with ReLU units builds its boundary from straight pieces in arbitrary directions, a tree from pieces parallel to the axes. For the smooth galaxies the boundaries are nearly lines of constant axis ratio, which a tree describes with a few questions.

To judge the tree we count its correct predictions on observations it has never seen. The accuracy is the fraction of correctly classified observations. When the classes have very different sizes, accuracy can be high for a model that ignores the small classes, since a model that never predicts cigar-shaped loses only 59 of 746 test galaxies. The balanced accuracy gives every class the same weight. It is the average over classes of the fraction of each class that is classified correctly,

$$\mathrm{BA} = \frac{1}{K} \sum_{k=1}^{K} \frac{T_k}{N_k}, \tag{4.10}$$

where N_k is the number of evaluation observations of class k and T_k the number of them classified correctly. The fraction T_k/N_k is called the recall of class k. A constant prediction of one class has $\mathrm{BA} = 1/K$ whatever the class sizes, and for classes of equal size the balanced accuracy equals the accuracy.

On the 853 validation galaxies the shallow tree reaches an accuracy of 0.8886 and a balanced accuracy of 0.8686. Its depth was fixed before any score was computed, so no selection took place, and the same tree enters the final comparison of Section 9 unchanged. Its test confusion matrix illustrates the arithmetic of equation (4.10). On the 746 test galaxies the tree classifies 348 of 398 round objects, 258 of 289 intermediate objects and 55 of 59 cigar-shaped objects correctly. The accuracy is $661/746 = 0.8861$, and the balanced accuracy is

$$\mathrm{BA} = \frac{1}{3}\left(\frac{348}{398} + \frac{258}{289} + \frac{55}{59}\right) = 0.8998. \tag{4.11}$$

The two scores differ because the classes have different sizes. Neither plays a part in choosing the questions, which minimise the Gini impurity of the fitting galaxies.

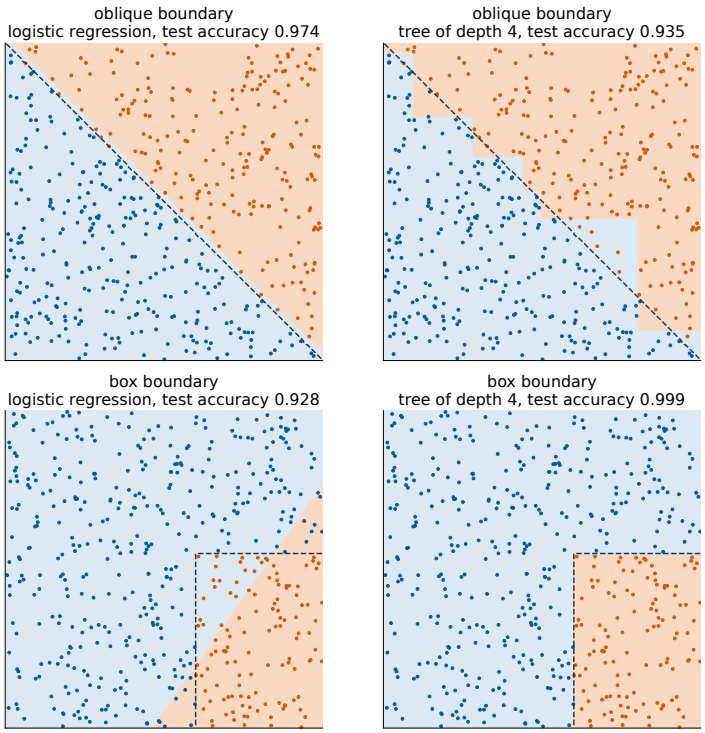


Figure 4.6: Logistic regression (left) and a tree of depth four (right) on two synthetic two-class tasks. Top, classes separated by an oblique line. Bottom, one class in an axis-parallel box. The dashed line is the true boundary, the shading the predicted class, and the titles give the accuracy on 5,000 test points.

Practice 3. Classes and probabilities

The rounder galaxy branch ends in leaves with round probabilities 76/109 and 1518/1544. Remove its last question and predict round everywhere in that branch. State which hard predictions change, which probabilities change and which kind of score can detect the change. Attempt this before reading the feedback in Appendix A.

4.6 Probability predictions and split geometry

The tree is now trained and we can explain why its rounder branch received another question even though both children predict round.

The Gini impurity also measures the error of the leaf probabilities. Encode the label of observation i as the indicator vector $\boldsymbol{e}_i$ with components $e_{ik} = \mathbf{1}\{y_i = k\}$, and let a node predict a probability vector $\boldsymbol{v}$. The mean squared distance between the indicator vectors and $\boldsymbol{v}$ splits into two terms,

$$\frac{1}{n_t} \sum_{i:\, x_i \in t} \|\boldsymbol{e}_i - \boldsymbol{v}\|^2 = I_G(t) + \|\boldsymbol{v} - \boldsymbol{p}_t\|^2, \tag{4.12}$$

where $\boldsymbol{p}_t = (p_{t1}, \ldots, p_{tK})$ is the mean of the indicator vectors. The cross term vanishes because the deviations $\boldsymbol{e}_i - \boldsymbol{p}_t$ sum to zero over the node, and $\frac{1}{n_t} \sum_i \|\boldsymbol{e}_i - \boldsymbol{p}_t\|^2 = \sum_k (p_{tk} - p_{tk}^2)$ is the Gini impurity, since each component $(\mathbf{1}\{y_i = k\} - p_{tk})^2$ averages over the node to $p_{tk} - 2p_{tk}^2 + p_{tk}^2$. Equation (4.12) has two consequences. The squared error of the predicted probabilities is smallest when $\boldsymbol{v}$ equals the class proportions, which justifies the leaf prediction (3.2), and the

smallest attainable error is $I_G(t)$. A node with low Gini impurity is a node whose labels a single probability vector predicts well.

For the Gini impurity the decrease takes a simple geometric form. Write $\boldsymbol{p}_t$, $\boldsymbol{p}_L$ and $\boldsymbol{p}_R$ for the proportion vectors of the parent and the children. Counting observations class by class gives $\boldsymbol{p}_t = \pi_L \boldsymbol{p}_L + \pi_R \boldsymbol{p}_R$. The constant ones in (4.1) cancel because $\pi_L + \pi_R = 1$, and

$$\begin{aligned} \Delta I_G &= \pi_L \|\boldsymbol{p}_L\|^2 + \pi_R \|\boldsymbol{p}_R\|^2 - \|\pi_L \boldsymbol{p}_L + \pi_R \boldsymbol{p}_R\|^2 \\ &= \pi_L(1-\pi_L)\|\boldsymbol{p}_L\|^2 + \pi_R(1-\pi_R)\|\boldsymbol{p}_R\|^2 - 2\pi_L\pi_R\, \boldsymbol{p}_L \cdot \boldsymbol{p}_R. \end{aligned} \tag{4.13}$$

Since $1 - \pi_L = \pi_R$ and $1 - \pi_R = \pi_L$, both coefficients in the second line equal $\pi_L \pi_R$, and the three terms combine into a square,

$$\Delta I_G = \pi_L \pi_R \, \|\boldsymbol{p}_L - \boldsymbol{p}_R\|^2 = \pi_L \pi_R \sum_{k=1}^{K} (p_{Lk} - p_{Rk})^2 \,. \tag{4.14}$$

The Gini decrease is never negative. It vanishes only when the two children have the same class proportions, and, holding the other factor fixed, it grows with the difference between the children or the balance of their sizes, since $\pi_L \pi_R$ is largest when $\pi_L = \pi_R = 1/2$. A question is valuable when knowing its answer changes the expected class composition, and a split that leaves the proportions unchanged has taught us nothing about the label.

4.7 Other impurity measures

Two other measures of mixing appear in the literature on trees. The misclassification error $I_M(t) = 1 - \max_k p_{tk}$ is the fraction of fitting observations that the majority class of the node misclassifies, and the entropy $I_E(t) = -\sum_k p_{tk} \log_2 p_{tk}$ is the average information of one label in bits. Both vanish for a pure node and peak at equal proportions. Appendix D shows that the misclassification error ignores every split whose two children keep the majority class of the parent, and that the entropy is the smallest mean log loss of a probability prediction in the same way as the Gini impurity is the smallest squared error. Our classification trees use the Gini impurity, the default of scikit-learn.

5 Regression trees

The galaxies required a category prediction. Many scientific targets are numbers, a reaction rate, a dose, a measured amount of disease progression, and a tree predicts them with the same questions and a different leaf value. Forests also combine numerical predictions, either regression outputs or class probabilities. The example is a clinical study of diabetes, where the target is a measured amount of disease progression.

In this study, ten quantities were recorded for each of 442 patients at the start of the study [6]. They are the age, the sex, the body mass index (BMI, the body mass in kilograms divided by the square of the height in metres), the average blood pressure in millimetres of mercury, and six measurements of the blood serum. One year later the physicians recorded for every patient a quantitative measure of how far the disease had progressed. Each patient is a row of a table. The ten baseline values form the input vector $x_i \in \mathbb{R}^{10}$ of patient i, and the progression score y_i is the target. The scores range from 25 to 346 and carry no physical unit, so we quote them as score units. The data are distributed with the scikit-learn library [7], from which we take them in their original units. Table 5.1 shows four patients with the two

Table 5.1: Four patients of the diabetes study. Row numbers refer to the order of the published table. BMI is in $\mathrm{kg\,m^{-2}}$, blood pressure in mm Hg.

Row	BMI	Average blood pressure	Progression score after one year
254	31.6	117	310
165	20.2	81	59
352	19.7	81	77
16	30.3	109	166

baseline measurements, BMI and average blood pressure, on which all later diabetes models are built.

We want to predict the progression one year later from the baseline measurements, and to see which measurements the prediction uses. The progression may depend on them in a nonlinear way, for example only once the BMI exceeds some value, and the effect of the blood pressure may be large for heavy patients and small for slim ones, an interaction between two inputs. A linear model assigns one fixed slope to every measurement and represents such an interaction only if the analyst writes it into the model in advance. A tree expresses it with the same kind of questions as for the galaxies. After a question about BMI, a question about blood pressure may be useful in one branch. Such a rule is a predictive pattern to be learned and assessed, and it carries no clinical recommendation.

We restrict the diabetes inputs to two measurements, BMI and average blood pressure. With two inputs every region of a fitted tree can be drawn in a plane, and every threshold can be read off. The methods apply unchanged to all ten measurements. As in the earlier lectures we divide the patients at random, with a fixed seed, into three sets. The 264 fitting observations (also called training observations) determine the thresholds and predictions of a model. The 89 validation observations are used to choose settings such as the depth of a tree. The 89 test observations assess models after validation-based selection. These course data are reused for several illustrations, and their scores serve as descriptive comparisons. The same three sets are used for every model, so that all scores refer to the same patients.

For these data the target is a number, and the tree must predict a progression score. The questions keep the form $x_j \le \theta$, and only two ingredients change, the value stored in a leaf and the measure of how well a node is predicted.

5.1 Leaf means and the squared-error criterion

A leaf ℓ with region R_ℓ predicts one number c for all observations in its region. Under squared error its cost is $S_\ell(c) = \sum_{i:\, x_i \in R_\ell} (y_i - c)^2$, where the sum runs over the n_ℓ fitting observations of the leaf. With the leaf mean $\bar{y}_\ell = n_\ell^{-1} \sum_{i:\, x_i \in R_\ell} y_i$, adding and subtracting $\bar{y}_\ell$ inside the square gives

$$S_\ell(c) = \sum_{i:\, x_i \in R_\ell} (y_i - \bar{y}_\ell)^2 + n_\ell \, (c - \bar{y}_\ell)^2, \tag{5.1}$$

since the cross term $2(\bar{y}_\ell - c) \sum_i (y_i - \bar{y}_\ell)$ vanishes. The first term does not depend on c, and the second is zero only at $c = \bar{y}_\ell$. The leaf therefore stores the mean response of its fitting observations, $c_\ell = \bar{y}_\ell$. Under absolute error the best constant would be the median, so the loss decides which summary a leaf stores. A regression tree with leaf regions $R_1, \dots, R_T$ and leaf means $c_1, \dots, c_T$ is the piecewise constant function

$$f(x) = \sum_{\ell=1}^{T} c_\ell \, \mathbf{1}\{x \in R_\ell\}. \tag{5.2}$$

The impurity of a regression node t is the remaining squared error, $\mathrm{SSE}(t) = \sum_{i \in t}(y_i - \bar{y}_t)^2$ with the node mean $\bar{y}_t$, and the score of a question is the decrease $\Delta\,\mathrm{SSE} = \mathrm{SSE}(t) - \mathrm{SSE}(L) - \mathrm{SSE}(R)$. Applying (5.1) to each child, with the parent mean in place of c, gives

$$\mathrm{SSE}(t) = \mathrm{SSE}(L) + \mathrm{SSE}(R) + n_L(\bar{y}_L - \bar{y}_t)^2 + n_R(\bar{y}_R - \bar{y}_t)^2. \tag{5.3}$$

Inserting $\bar{y}_t = \pi_L \bar{y}_L + \pi_R \bar{y}_R$, so that $\bar{y}_L - \bar{y}_t = \pi_R(\bar{y}_L - \bar{y}_R)$ and $\bar{y}_R - \bar{y}_t = -\pi_L(\bar{y}_L - \bar{y}_R)$, and using $n_L = n_t \pi_L$, $n_R = n_t \pi_R$, so that $n_L \pi_R^2 + n_R \pi_L^2 = n_t \pi_L \pi_R(\pi_R + \pi_L) = n_L n_R / n_t$, the decrease becomes

$$\Delta\,\mathrm{SSE} = \left(n_L \pi_R^2 + n_R \pi_L^2\right)(\bar{y}_L - \bar{y}_R)^2 = \frac{n_L n_R}{n_t}(\bar{y}_L - \bar{y}_R)^2. \tag{5.4}$$

The structure is that of (4.14). A split earns a large decrease when two large groups have different mean responses. Dividing by n_t gives the decrease in the mean squared error of the node, $\Delta\,\mathrm{SSE}\,/n_t$, which is the local decrease quoted for the diabetes tree. The node impurity stored by scikit-learn is its MSE. The parameter `min_impurity_decrease` uses the globally weighted reduction $(n_t/N)(\Delta\,\mathrm{SSE}\,/n_t) = \Delta\,\mathrm{SSE}\,/N$, where N is the total fitting size. At a fixed node both normalizations rank candidate questions identically. Figure 5.1 shows the criterion on three responses $y = (1, 2, 6)$ at inputs $x = 1, 2, 3$. The parent has mean 3 and $\mathrm{SSE} = 14$. The threshold 1.5 separates the smallest response and leaves $\mathrm{SSE} = 8$, a decrease of 6. The threshold 2.5 separates the largest response and leaves $\mathrm{SSE} = 0.5$, a decrease of 13.5, in agreement with $\frac{2 \cdot 1}{3}(1.5 - 6)^2$.

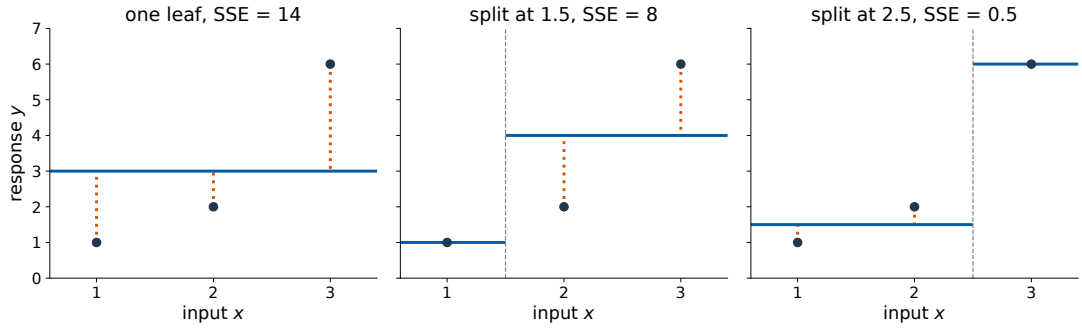


Figure 5.1: Three responses with one leaf (left) and with the two possible splits (centre and right). Blue segments are the leaf means and dotted lines the residuals. Grouping the two small responses leaves the smaller squared error.

Practice 4. A numerical target

The fitting responses at inputs $1, 2, 3$ are $1, 2, 6$. Calculate the parent mean, compare thresholds 1.5 and 2.5 by total child squared error, and predict the response at input 2.2 using the winning split. Attempt this before reading the feedback in Appendix A.

One constant prediction in each node

Every node summarizes its fitting targets by a constant prediction chosen to minimize a loss. A split replaces that constant by two constants, one in each child. The remaining minimum average loss is the impurity.

Target and loss	Best constant	Node impurity
Number, squared error	Mean response	MSE
Class indicator vector, squared error	Class proportions	Gini
Class label, log loss	Class proportions	Entropy

For classification, a class indicator vector has one in the observed class and zero in every other component. Equation (4.12) derives the Gini row. Appendix D derives the entropy row. Entropy and log loss must use the same logarithm base.
The regression impurity is $\mathrm{MSE}(t) = \mathrm{SSE}(t)/n_t$, with divisor n_t. The sample variance used to estimate a population variance has divisor $n_t - 1$ and serves a different purpose. Minimizing total child SSE and minimizing size-weighted child MSE choose the same split within a fixed parent.

5.2 A regression tree for diabetes progression

The diabetes tree is grown on the 264 fitting patients with the two inputs BMI and average blood pressure, a maximum depth of two and at least ten patients per leaf. Figure 5.2 shows the fitting data. The progression tends to increase with BMI, and at a given BMI it scatters over more than a hundred score units.

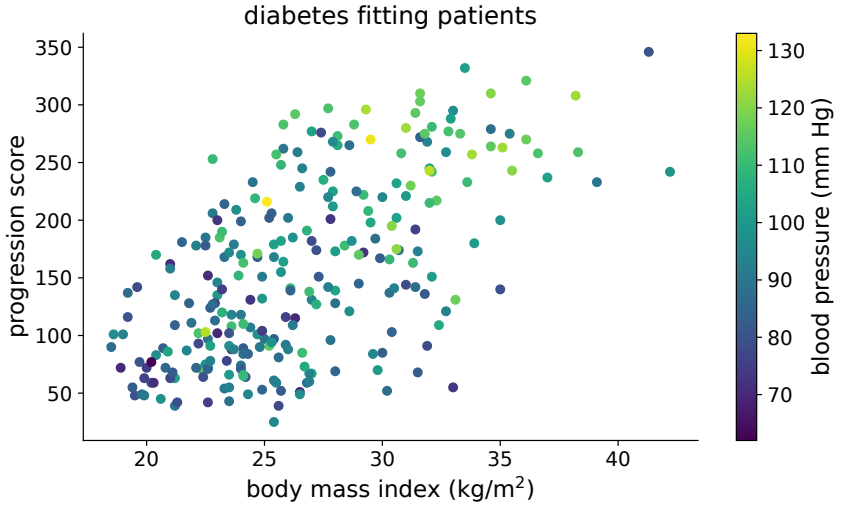


Figure 5.2: The 264 fitting patients of the diabetes study. The horizontal axis is BMI, the colour the average blood pressure and the vertical axis the progression score after one year.

At the root the 264 fitting patients have a mean score of 155.6, and the mean squared deviation of their scores from this mean is 5998.6 squared score units. The best question is BMI ≤ 27.35, which sends 162 patients to the left and 102 to the right and lowers the weighted mean squared error of the children to 4109.7. The search then continues in each child separately. Table 5.2 lists the three questions of the fitted tree, and Figure 5.3 shows the tree.

The two children ask different questions. Among the slimmer patients the tree asks about BMI again, at 22.75, and the leaf means are 94.5 and 134.4. Among the heavier patients it asks about blood pressure, and the two leaves of 51 patients each have means 177.2 and 243.5. The tree has found an interaction. Blood pressure enters the prediction only for patients with BMI greater than 27.35, where it changes the predicted progression by 66 score units, and a model with one global slope per measurement cannot express this. The first validation

Table 5.2: The three questions of the depth-two diabetes regression tree. MSE values are in squared score units, and the last column is the decrease $\Delta\,\mathrm{SSE}\,/n_t$ of the node mean squared error.

Node	Question	n_t	n_L/n_R	Node MSE	Weighted child MSE	Decrease
root	BMI $\leq$ 27.35	264	162/102	5998.63	4109.66	1888.97
left	BMI $\leq$ 22.75	162	54/108	3612.39	3259.79	352.60
right	blood pressure $\leq$ 101.5	102	51/51	4899.44	3800.07	1099.38

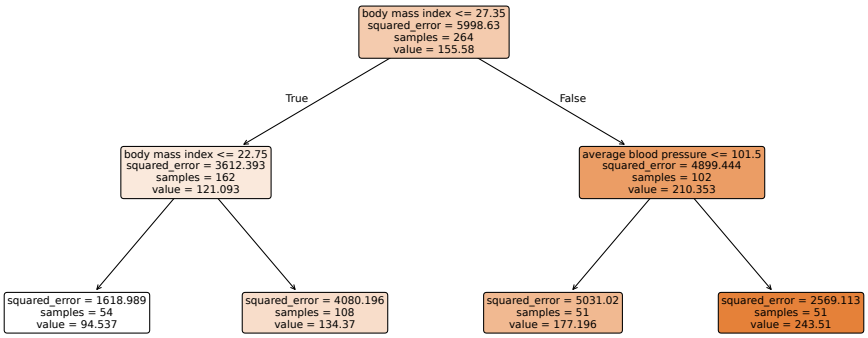


Figure 5.3: The depth-two regression tree for diabetes progression. Each box gives the question (pressure stands for average blood pressure), the number n of fitting patients and their mean progression score. The mean of a leaf is its prediction.

patient, row 6, has BMI 22.0 and answers yes at both BMI questions. The tree predicts the mean of the 54 fitting patients in that leaf, 94.5, while the recorded progression of this patient is 138. The sample standard deviation of the scores within the leaf is 40.6, so a deviation of this size is common in the region. The explanation of a prediction by its path is complete, and the prediction can still be far from the observation.

Figure 5.4 draws the same tree in the plane of the two measurements, in the manner of the baseball-salary example of Section 8.1.1 of [2]. The four leaves are four rectangles, and the tree predicts the leaf mean everywhere inside a rectangle. The root question draws the vertical line at BMI 27.35. The second BMI question cuts only the left half and the blood-pressure question only the right half, so the line at 101.5 mm Hg stops at BMI 27.35. The interaction is visible as this geometry. The effect of blood pressure on the prediction depends on the value of BMI, and the tree represents this dependence by asking different questions in different branches.

5.3 Error measures

For n evaluation observations with predictions $\widehat{y}_i$, the mean squared error and the root mean squared error are

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}. \tag{5.5}$$

The MSE has squared response units, and the RMSE has the units of the response and can be read as a typical prediction error. On the 89 test patients the depth-two tree has RMSE = 69.0 score units. A constant prediction equal to the mean fitting score gives 79.2. The two baseline measurements explain part of the variation of the progression, The remaining prediction error

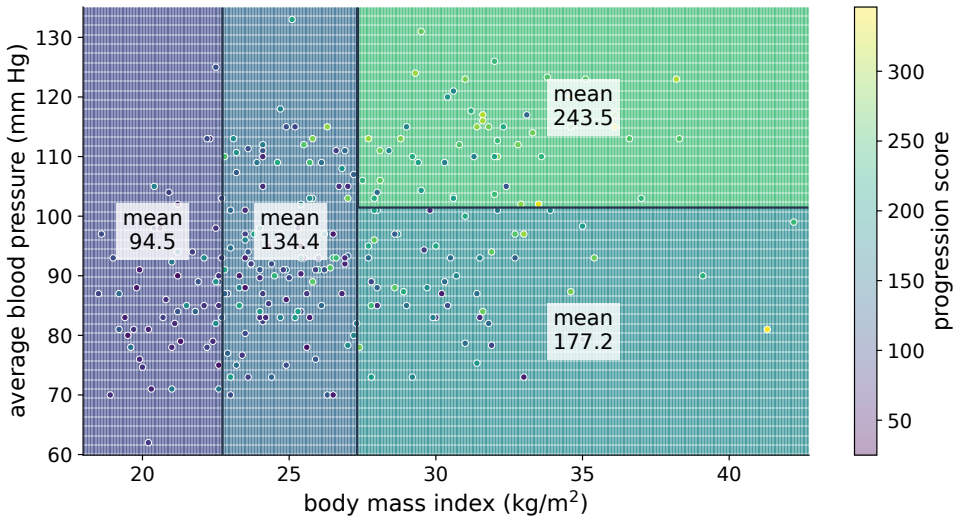


Figure 5.4: The depth-two diabetes tree as four rectangles in the plane of BMI and average blood pressure, coloured by the mean progression score of the fitting patients in each leaf (94.5, 134.4, 177.2 and 243.5). The dots are the fitting patients, coloured by their scores.

can include omitted measurements, measurement noise and limitations of this fitted model. This comparison alone cannot separate their contributions. Because every leaf value is a mean of fitting responses, a regression tree cannot predict beyond the range of the responses it was fitted to, a limitation shown on synthetic data with a known law in Section 9.4. A supplementary notebook turns the same patients' scores into two classes using the fitting median.

6 Tree complexity and validation

A tree can grow until every leaf is pure or holds a single observation. Such a tree reproduces the fitting labels, including their noise, and its leaves contain too few observations to estimate a probability or a mean. A tree that is too shallow assigns one prediction to regions in which the target changes. The maximum depth and the minimum number of observations per leaf control the balance between these two failures. They are hyperparameters, settings chosen outside the threshold search, and they are chosen by comparing candidate settings on the validation observations, which play no part in growing the trees. The test observations are kept for the end, to describe the chosen model. We return to the galaxies, where the classes are large enough for these choices to be visible. We control depth directly and choose it on validation observations. This keeps the student workflow visible. More advanced complexity-selection methods are described in the cited textbooks [1, 2].

6.1 The full descriptor row

The depth-two tree saw only the axis ratio, the concentration and the failure flag. The same masked image yields further numbers, and a deeper tree can use them. We give every galaxy a row of eight entries, all computed from the 64-pixel image without the label. The mask area fraction is the fraction of the 64×64 pixels inside the mask. The brightness sum is the total background-subtracted display brightness W of (B.1), divided by 255. The half-light radius r_{50} is the radius in pixels that encloses half of W. The axis ratio q and the concentration C are

those of (B.2). The red-blue index compares the background-subtracted red and blue display channels summed over the mask. In the Galaxy Zoo DECaLS images the red channel carries the z-band and the blue channel the g-band image [5], so the index compares a near-infrared with a green-blue brightness, $(R-B)/(R+B)$, so that redder objects have larger values. The centre offset is the distance in pixels between the brightness-weighted centre $\bar{\boldsymbol{r}}$ and the centre of the image, and the last entry is the mask-failure flag. Missing values of the one failed fitting mask are again replaced by fitting medians.

The new descriptors are strongly related to each other. Figure 6.1 gives the Pearson correlation coefficients over the fitting galaxies. The mask area, the brightness sum and the half-light radius all measure the apparent size of the galaxy, and their pairwise correlations lie between 0.93 and 0.97. The axis ratio and the concentration have correlation -0.65. The radii r_{50} and r_{90} are radii of circles, and a circle drawn around an elongated image has to grow far along the major axis before it encloses 90 percent of the light, so a small q raises r_{90}/r_{50} for geometric reasons. Correlated descriptors can substitute for each other in the search, changing the chosen questions and sometimes predictions. They also affect how credit is shared between descriptors, which becomes important in Section 7.

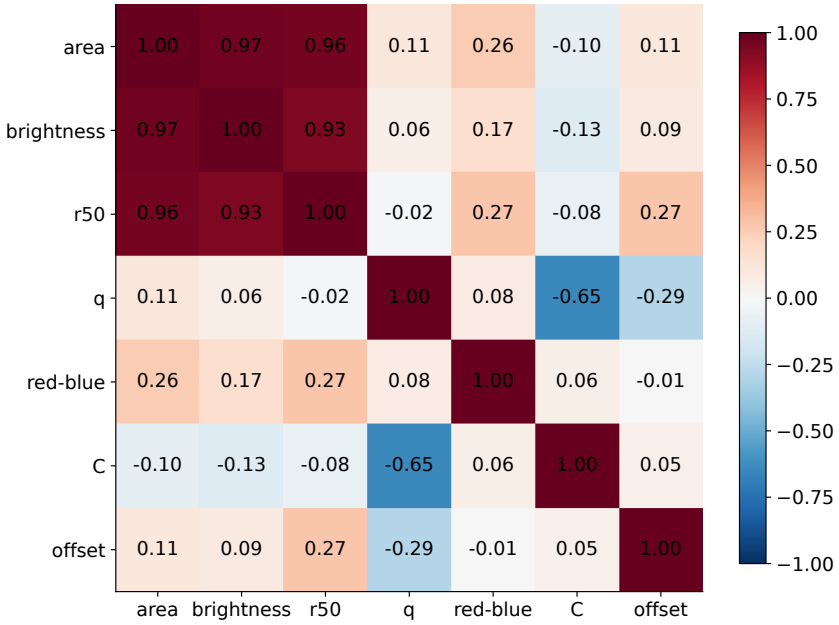


Figure 6.1: Pearson correlation coefficients between the seven numerical descriptors over the 3,407 fitting galaxies. Mask area, brightness sum and half-light radius form a block of nearly equivalent size measures.

Grown to depth two on the full row, the tree asks the same three questions on q with the same thresholds as in Section 4. No other descriptor reduces the Gini impurity at these nodes by more than the axis ratio does.

Practice 5. A decision before testing

Write a rule for selecting a depth using only fitting and validation observations. Include a tie rule and a possible preference for a shorter explanation. Freeze this rule before examining the test scores. Explain the distinct roles of the Gini criterion and balanced accuracy. Attempt this before reading the feedback in Appendix A.

6.2 Depth selection on the galaxies

We grow trees of maximum depth 1 to 15 on the 3,407 fitting galaxies, with at least ten galaxies per leaf, and score each on the fitting and on the 853 validation galaxies with the balanced accuracy (4.10). Figure 6.2 shows the two curves. A single question on q reaches 0.61. The second level of questions lifts both curves to almost 0.89 and 0.87, because it separates the cigar-shaped galaxies from the intermediate ones. Beyond depth two the fitting score creeps up to 0.91 while the validation score stays between 0.852 and 0.874. The deeper trees describe the fitting galaxies a little better and new galaxies no better. From depth eight onwards the fitting balanced accuracy stays at 0.9119, although the tree keeps growing, to 80 leaves at depth 13.

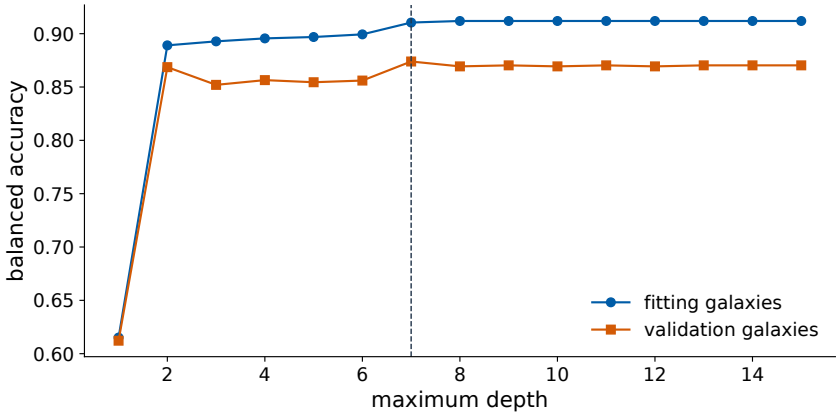


Figure 6.2: Fitting and validation balanced accuracy of galaxy trees with at least ten galaxies per leaf against the maximum depth. The dashed line marks depth seven, the largest validation score.

Table 6.1: Selected galaxy trees. Depth seven has the largest validation balanced accuracy, and depth two is the tree of Section 4. Test scores refer to the 746 test galaxies.

Maximum depth	Leaves	Validation BA	Test accuracy	Test BA
2	4	0.8686	0.8861	0.8998
7	55	0.8739	0.9169	0.8754

The validation curve peaks at depth seven, a tree of 55 leaves, and Table 6.1 compares it with the depth-two tree. The two validation scores differ by 0.005. Before evaluation we fix the candidate depths, the minimum leaf size and the rule of choosing the largest validation balanced accuracy, with ties going to the smallest depth. We then freeze the selected tree and inspect its test scores. One of the 55 cigar-shaped validation galaxies moves the balanced accuracy by $1/(3 \cdot 55) = 0.006$, so the difference is smaller than the effect of a single galaxy of the rarest class. This calculation describes score granularity. An uncertainty estimate would need a sampling calculation, such as resampling evaluation galaxies within each class for fixed models. Repeated fitting and selection require a separate experiment with new fitting splits. On the test galaxies the deeper tree has the higher accuracy and the lower balanced accuracy. A confusion matrix shows where the difference comes from. Its entry in row k and column l counts the evaluation observations of class k that the model assigns to class l, so the diagonal holds the correct predictions and the recall of class k is the diagonal entry divided by the row sum. Figure 6.3 gives the matrix of the depth-seven tree. It classifies 381 of 398 round galaxies correctly, against 348 for the depth-two tree, and 46 of 59 cigar-shaped galaxies, against 55. The many round galaxies dominate the accuracy, the few cigar-shaped galaxies carry a third of the balanced accuracy, and the two scores move in opposite directions. The

deeper tree is preferred by the validation rule, the shallow tree is about as good and can be read in a glance. A preference for the simpler model can be specified before testing, for example by requiring a stated validation improvement to justify more leaves. In this analysis the declared maximum-validation rule selects depth seven. The depth-two tree remains a separate, fixed teaching model. Its favourable test score does not retrospectively change that selection rule. The shorter regression applications use fixed configurations and compare their validation scores before testing.

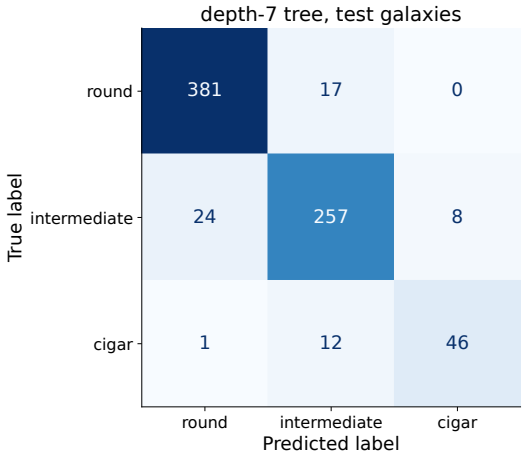


Figure 6.3: Confusion matrix of the depth-seven tree on the 746 test galaxies. Rows are catalogue classes, columns predicted classes.

7 Interpreting a single tree

A tree is called interpretable because every prediction can be traced. Interpretation of a fitted model nevertheless asks two different questions, and each needs its own tool. The first concerns one prediction and is answered by its decision path. The second asks which features the model uses across all observations, and two importance measures answer it in different ways. Every answer describes the fitted model and its data, and none of them is a statement about cause and effect in the galaxies.

7.1 Decision paths

The prediction of a tree for one input is determined by its path, the sequence of questions and answers that leads to a leaf, and by the fitting observations in that leaf. The region of the leaf is the intersection (4.9) of the half-spaces selected along the path. The galaxy of row 4234 follows two questions about q and reaches the class fractions in (4.8). Neither C nor any other descriptor affects this prediction of the shallow tree. In the depth-seven tree the same galaxy answers up to seven questions, and a reader can still check each of them, although the path is no longer a rule one remembers. A descriptor that does not appear on a path can still matter for other galaxies, whose paths ask about it. A path describes one prediction and says nothing about the role of a feature in the model as a whole.

7.2 Impurity importance

Each internal node t of a tree lowers the impurity of the n_t observations that reach it by $\Delta I(t)$, the decrease (4.4), or for regression the node mean squared error decrease $\Delta\,\mathrm{SSE}\,/n_t$. Weighting each decrease by the fraction n_t/N of fitting observations it affects and summing over the nodes that use feature j gives the impurity importance, also called the mean decrease in impurity [1],

$$A_j = \sum_{t:\ j_t=j} \frac{n_t}{N}\,\Delta I(t), \qquad \mathrm{Imp}_j = \frac{A_j}{\sum_{l=1}^{d} A_l}, \tag{7.1}$$

where j_t is the feature of node t. The weighted decreases of all nodes telescope. Each node contributes $(n_t/N)I(t)$ minus the corresponding terms of its children, the child terms cancel against the contributions of the children as parents, and $\sum_l A_l$ equals the impurity of the root minus the size-weighted impurity of the leaves. Imp_j is the share of this total decrease achieved by questions on feature j.

All splits of the depth-two galaxy tree use q, so its normalised impurity importance is one and every other descriptor receives zero. The zero describes this fitted tree. The concentration does carry information about the class, as its root decrease of 0.084 in Table 4.1 shows, but at every node of this tree q offered more. If a tree has no positive total impurity decrease, the normalisation in (7.1) is undefined and implementations report zeros. Impurity importance is computed from the fitting observations. It favours features with many distinct values, which offer more thresholds and hence more chances of a decrease by chance, and it divides the credit between correlated features according to which of them happened to win the splits.

The depth-seven galaxy tree uses all seven numerical descriptors. Table 7.1 and Figure 7.1 give its impurity importance. The axis ratio collects 0.83 of the total decrease, and the centre offset is second with 0.12. The size measures, the colour and the concentration share the remaining five percent, through questions deep in the tree that act on few galaxies.

Table 7.1: Impurity importance (fitting galaxies) and permutation importance (mean drop of the validation balanced accuracy over ten shuffles in percentage points, with its standard deviation over the shuffles) of the depth-seven galaxy tree.

Descriptor	Impurity importance	Permutation drop (points)	Shuffle s.d. (points)
mask area fraction	0.0075	0.25	0.10
brightness sum	0.0106	0.76	0.20
half-light radius r_{50}	0.0059	0.19	0.12
axis ratio q	0.8284	54.06	1.50
red-blue index	0.0115	0.72	0.25
concentration C	0.0130	0.33	0.40
centre offset	0.1232	3.99	1.01
mask failure	0.0000	0.00	0.00

7.3 Permutation importance

A different question is how much the predictions of a fitted model depend on a feature. Breiman answered it by destroying the information in the feature and measuring the damage [8]. Breiman defined it for forests, which Section 8 introduces. We use the validation data, which serve every kind of model. Take a score s for which larger is better, such as the balanced accuracy or the negative RMSE, and the validation data $\mathcal{D}_{\mathrm{val}}$. Shuffle the values of feature j among the validation observations, which keeps its distribution and breaks its relation to the

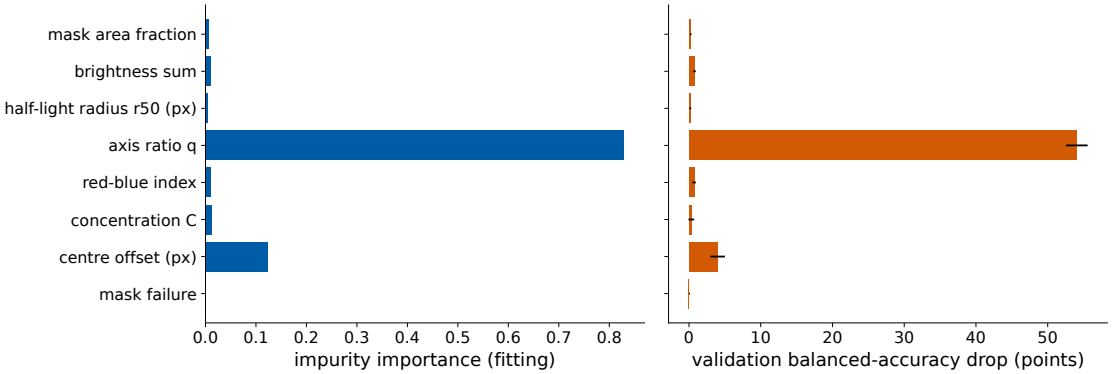


Figure 7.1: Impurity importance (left) and validation permutation importance in percentage points of balanced accuracy (right) for the depth-seven galaxy tree. Error bars give the standard deviation over ten shuffles.

target and to the other features, and score the unchanged model again. Repeating the shuffle n_{sh} times, with shuffles $\kappa = 1, \ldots, n_{\mathrm{sh}}$,

$$\mathrm{PI}_j = \frac{1}{n_{\mathrm{sh}}} \sum_{\kappa=1}^{n_{\mathrm{sh}}} \left[s(f, \mathcal{D}_{\mathrm{val}}) - s(f, \mathcal{D}_{\mathrm{val}}^{(j,\kappa)}) \right], \tag{7.2}$$

where $\mathcal{D}_{\mathrm{val}}^{(j,\kappa)}$ is the validation set after shuffle κ of feature j. The model is never refitted. For the galaxies we use $n_{\mathrm{sh}} = 10$ shuffles. The spread over the shuffles describes the randomness of the shuffling alone and gives no uncertainty of the importance with respect to new data. PI_j has the units of the score, a decrease of balanced accuracy or an increase of RMSE. We quote drops of balanced accuracy in percentage points, so a drop from 0.87 to 0.33 is a drop of 54 points.

For the depth-seven galaxy tree, shuffling q lowers the validation balanced accuracy by 54 points, from 0.87 to about a third, the value of a constant prediction. Without its axis ratio the tree is reduced to guessing. Shuffling the centre offset costs 4.0 points, and every other descriptor costs less than one point (Table 7.1). The two importance measures agree on the order of the first two descriptors and disagree on the size of the gap, because the impurity importance counts decreases on the fitting galaxies and the permutation importance counts damage to predictions on new galaxies.

The two importances can also disagree about a single question. The right-hand question $q \leq 0.823698$ of the depth-two galaxy tree sends the 1,653 rounder galaxies into two leaves that both predict round (Section 4.4). It lowers the Gini impurity of its node by 0.010093, and by (7.1) it adds

$$\frac{1653}{3407} \cdot 0.010093 = 0.0049 \tag{7.3}$$

to the impurity importance A_q of the axis ratio. No galaxy changes its predicted class when it answers this question differently, so the question contributes nothing to the accuracy, to the balanced accuracy or to any permutation importance computed from hard predictions. A score computed from the probabilities, such as the log loss, does register it, because the round probability of the two leaves is $76/109 = 0.70$ in one and $1518/1544 = 0.98$ in the other. In the depth-two tree the effect is hidden behind the large importance of q at the root.

Two limitations of permutation importance follow from its construction. When two features carry overlapping information, a model that uses both can lose little when one of them is shuffled, and both may appear unimportant. Shuffling one feature independently of the others

also creates combinations of values that never occur, such as a small mask area with a large brightness sum, and the model is then scored on inputs unlike any it was fitted to.

7.4 Measurement artefacts in the descriptors

The centre offset is the second most important descriptor of the galaxy tree, and it describes the image and its processing more than the galaxy. A galaxy catalogued at the centre of its cut-out has a small offset. A large offset arises when the mask includes a neighbouring star or galaxy, or when the object is so faint or so elongated that the brightness-weighted centre drifts from the image centre. Large offsets are more frequent among the round fitting galaxies, as Table 7.2 shows, and Figure 7.2 shows the reason. In each of the six round galaxies with the largest offsets the mask has merged the galaxy with a bright neighbour. The light-weighted centre lies between the two objects, and the axis ratios between 0.30 and 0.58 describe the elongated pair, although each galaxy is round. Without the offset such a galaxy would land in the leaves of elongated galaxies. The use of the offset by the tree is consistent with this picture, a large offset marking galaxies whose q is unreliable, although the six images and the class frequencies do not prove that this is the only reason for the questions on the offset. The descriptor carries information about the measurement procedure, and a model that relies on it works only as long as new images are prepared in the same way. Importances alone cannot tell whether a descriptor carries physics or an artefact of the measurement. That judgement needs knowledge of how the data were made, and the images of the galaxies concerned.

Table 7.2: Centre offset of the fitting galaxies of each class.

class	median offset (px)	fraction above 2 px	fraction above 5 px
round smooth	0.343	0.096	0.021
intermediate smooth	0.337	0.065	0.015
cigar-shaped smooth	0.414	0.050	0.009

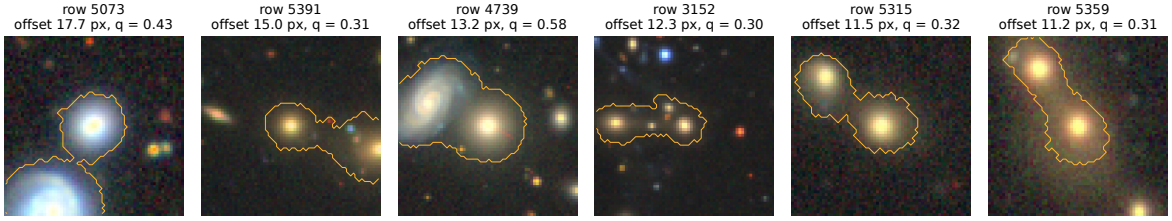


Figure 7.2: The six round fitting galaxies with the largest centre offsets, with their masks. Each mask includes a neighbouring object, which moves the light-weighted centre away from the galaxy and lowers its axis ratio.

Practice 6. A scientific explanation

The centre offset is important to the fitted model. Separate the observed evidence, the proposed masking explanation and a test of that explanation. Describe a comparison after correcting contaminated masks, with the evaluation observations fixed. Attempt this before reading the feedback in Appendix A.

8 Random forests

A single tree supplies a short explanation when its depth is small. A larger tree can capture more detail, but its questions can change when the fitting observations change. A random forest combines the predictions of many randomized trees [8]. Scikit-learn handles the randomization and fitting internally. We use the forest as another estimator with the same `fit` and `predict` interface.

For regression the forest averages its trees' numerical predictions. For classification it averages their class-probability vectors and returns the class with the largest average probability,

$$\widehat{f}(x) = \frac{1}{B}\sum_{b=1}^{B} f_b(x), \qquad \widehat{p}_k(x) = \frac{1}{B}\sum_{b=1}^{B} \widehat{p}_{b,k}(x), \qquad \widehat{y}(x) = \arg\max_k \widehat{p}_k(x). \tag{8.1}$$

Here B is the number of trees and $\widehat{p}_{b,k}$ is the probability returned by tree b. Averaging probabilities can give a different decision from voting on hard class predictions. For example, two trees predicting $(0.51, 0.49)$ and $(0.01, 0.99)$ have mean $(0.26, 0.74)$, while their hard votes tie.

Using a forest

`RandomForestClassifier` predicts classes and probabilities. `RandomForestRegressor` predicts numbers. The parameter `n_estimators` sets the number of trees. A larger value costs more computation and can make the combined prediction steadier. The parameter `min_samples_leaf` controls how many fitting observations support a leaf. We begin with a small number of declared settings and compare them on validation observations when selection is needed.
The fitting procedure is handled by the library. The student workflow remains fitting, validation, a frozen choice and test evaluation. Increasing the number of trees gives no guarantee that every successive hard-decision score improves.

A forest is usually harder to summarize than a short tree because a prediction combines many paths. We can still inspect its predictions and apply permutation importance. Agreement among trees measures agreement within this fitted model. Shared measurement errors can make the trees agree on an incorrect class. Probability reliability and predictive accuracy need held-out evidence.

Practice 7. Combining predictions

Average the probability vectors $(0.51, 0.49)$ and $(0.01, 0.99)$ and compare their decision with hard voting. Explain why unanimous forest predictions can still be wrong. Feedback is in Appendix A.

9 Comparing the methods in applications

We now use a common workflow for a decision tree, random forest and small neural network. Each comparison supplies the same inputs and fitting, validation and test observations to the three methods. A classifier returns a class with `predict` and a probability vector with `predict_proba`. A regressor returns a number with `predict`. The notebooks expose these scikit-learn calls directly.

9.1 A common fitting and evaluation procedure

The fitting observations teach every model. A pipeline estimates preprocessing statistics from those observations and applies the stored transformation to validation and test inputs. The neural network benefits from standardizing input columns with their fitting means and standard deviations. A regression network also benefits from target scaling. Scikit-learn's `TransformedTargetRegressor` performs that scaling and converts predictions back to response units. Trees can consume measurements in their original units.

We declare model settings before test inspection. A short application can use fixed settings and inspect their validation scores. If alternatives are compared, validation chooses among them. We then freeze the procedure and evaluate the test observations once. A constant class or mean-response prediction provides a baseline. A confusion matrix locates classification errors, while a plot of predicted against observed responses locates regression errors. Similar average scores can conceal different errors on individual observations.

9.2 Galaxy measurements

The galaxy application first supplies eight measured descriptors to all three methods. The short tree from Section 4 remains a readable reference. A depth-seven tree, a forest and a network provide the richer comparison. These configurations are stated in the notebook, and the depth is selected on validation observations. The table reports the frozen models on the same test objects.

Model	Test accuracy	Test BA
Tree	0.917	0.875
Forest	0.932	0.902
Network	0.948	0.937

These scores describe this representation, fixed configurations and reused course split. The network receives measured descriptors in this comparison. Learning from image pixels is a separate experiment below. The centre-offset example of Section 7.4 remains a reason to inspect the measurements used by every model.

9.3 Classification directly from images

The MNIST application receives 784 brightness values per image, obtained by dividing each 28×28 pixel array by 255 and flattening its rows. All three methods receive those same inputs. A tree question compares one pixel with a threshold. A dense neural network combines pixel values in its hidden layer. A forest combines its fitted trees internally. The notebook compares three tree depths on validation images, then fits a forest and a network with declared settings. Internal network early stopping uses a subset of fitting images. The external validation and test images remain outside training.

Model	Test accuracy
Baseline	0.114
Tree	0.822
Forest	0.951
Network	0.942



Figure 9.1: The same twelve test images with predictions from the tree, forest and network, denoted T, F and N. The images are the first twelve test rows, chosen independently of model success. A disagreement identifies an object worth inspecting.

The galaxy application also receives pixels directly. We average each 4×4 block of a 64×64 colour image to obtain $16 \times 16 \times 3$ brightness values, scale them to $[0, 1]$ and flatten the result to 768 inputs. Every method sees this same reduced image and the same three smooth classes. Reduction keeps the CPU example small and can discard fine structure. This experiment has a different representation from the descriptor comparison.

Model	Test accuracy	Test BA
Tree	0.798	0.683
Forest	0.877	0.737
Network	0.879	0.678

The galaxy pixel comparison also illustrates why accuracy and balanced accuracy can rank models differently. The network has the largest accuracy, while the forest has the largest balanced accuracy. Inspect the class recalls to locate the difference. The rare cigar-shaped class contributes equally to balanced accuracy and little to total accuracy.

All three image classifiers can be affected by changes of object position because the input coordinates refer to fixed pixels. A convolutional network introduces spatial weight sharing, which is a different architecture. These small comparisons illustrate the methods on identical inputs and do not establish a general ranking of image classifiers.

9.4 Numerical regression

The diabetes notebook compares the same model families on numerical progression using BMI and average blood pressure. RMSE expresses error in score units and is compared with the

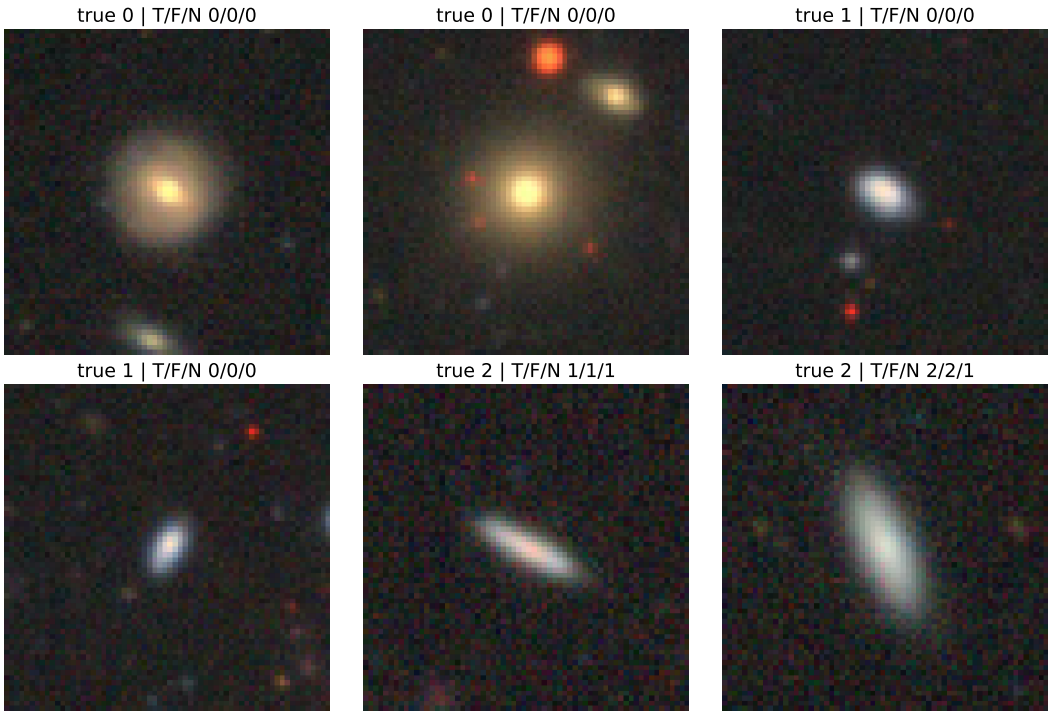


Figure 9.2: Six test galaxies with tree, forest and network predictions, the first two test rows of each class chosen independently of model success. Class codes 0, 1 and 2 mean round, intermediate and cigar-shaped. The displayed originals show the objects, while every model receives the same reduced 16×16 RGB input.

constant fitting-mean baseline. The tree uses depth two and at least ten observations per leaf. The forest and the network use fixed teaching configurations.

Model	Test RMSE
Baseline	79.2
Tree	69.0
Forest	72.2
Network	67.5

The concentration-response application gives a simpler one-dimensional view. We generate noisy rates around $r(s) = 2.4s/(1.2 + s)$ for concentrations s between 0.1 and 4, fit each estimator using only the measured pairs and display its predictions. The generating curve is available to the reader but is never supplied as a model to fit. The tree predicts one mean response per interval. The forest combines such predictions. The network gives another fitted dependence on concentration.

A regression tree's leaf means remain within the range of its fitting responses. A forest averages such values and inherits this bound. With one input both become constant beyond their outer thresholds. A network can continue changing outside the measured range, but that continuation is determined by its fitted function and requires separate evidence. A low interpolation error alone cannot validate extrapolation.

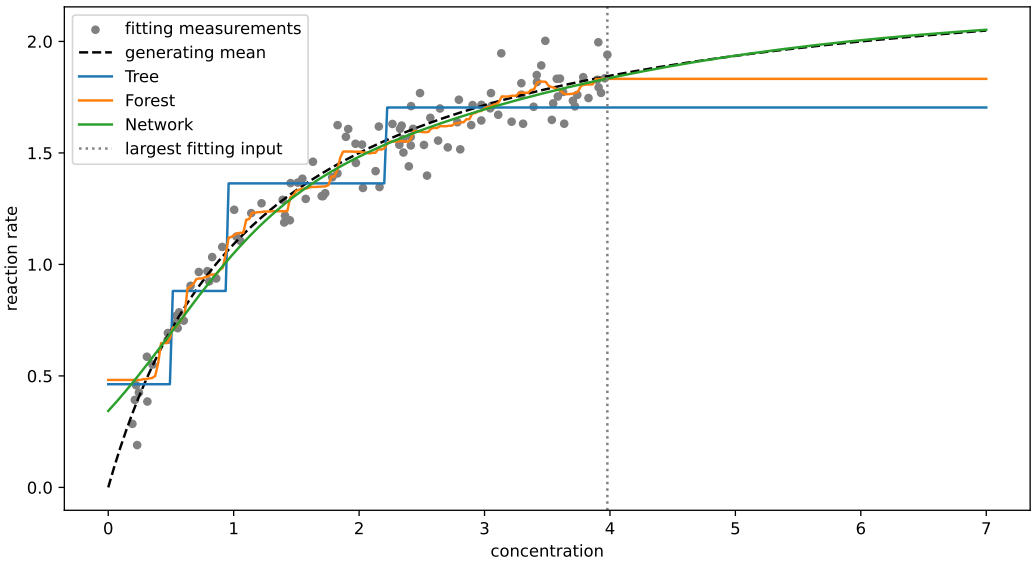


Figure 9.3: Tree, forest and network predictions for concentration-response measurements. The dashed curve generated the noisy observations. The vertical line marks the largest fitting concentration. Test error within the measured range and behaviour beyond that range are separate questions.

Practice 8. Comparing methods

Choose a displayed image on which the methods disagree and inspect their probabilities. For regression, identify a response with a large prediction error. State what the shared inputs and split let us compare, and what a single test score cannot establish. The short notebooks keep the workflow visible without implementing a learning algorithm.

A Practice feedback

1. Mixing in a node

The proportions are $(0.8, 0.2)$ and the hard prediction is A. Its error fraction is 0.2. Two independent labels disagree with probability $2(0.8)(0.2) = 0.32$. These probabilities describe different experiments.

2. Comparing questions

The parent Gini impurity is 0.48. U leaves weighted impurity $0.5(0.32) + 0.5(0.48) = 0.40$ and gains 0.08. V leaves $0.9[1 - (5/9)^2 - (4/9)^2] = 4/9$ and gains approximately 0.036. U wins. A candidate needs only the feature values and fitting labels reaching this node, plus the fixed child-size rule. Validation labels play no part.

3. Classes and probabilities

Both original leaves and their parent predict round, so every hard prediction in this branch is unchanged. The parent predicts round with probability 1594/1653. The split replaces that probability with 76/109 or 1518/1544. Log loss and squared probability loss can register this

change. Accuracy and balanced accuracy cannot. A fitting improvement does not guarantee a validation improvement.

4. A numerical target

The mean is 3 and parent SSE is 14. Threshold 1.5 gives means 1 and 4, with child SSE sum 8. Threshold 2.5 gives means 1.5 and 6, with child SSE sum 0.5. The latter wins and predicts 1.5 at 2.2. The prediction is constant in the reached interval.

5. A decision before testing

One valid rule chooses the largest validation balanced accuracy, breaking ties by smaller depth. Another declares in advance a minimum validation improvement needed to justify extra leaves. The preference is part of the procedure and must precede test inspection. Gini selects fitting questions locally. Balanced accuracy ranks the resulting candidate models on validation observations. Test observations assess the frozen procedure.

6. A scientific explanation

The importance measures establish model reliance on offset. The inspected images show contaminated masks in six extreme cases. The proposed explanation is that offset helps identify distorted axis ratios. Correct masks using a procedure chosen without evaluation labels, recompute descriptors and compare predictions before and after correction on the same evaluation objects. Compare performance and offset reliance after refitting with the corrected representation. This tests a measurement explanation. It does not establish a causal effect of physical galaxy size or shape.

7. Averaging and agreement

The average is $(0.26, 0.74)$ and predicts the second class. Hard votes tie because each tree prefers a different class. Unanimity establishes agreement among the fitted trees. Accuracy on new observations and calibration of their probabilities require held-out evidence. Shared measurement errors can make all trees agree on a wrong answer.

8. Comparing methods

The common inputs and split let us compare these fitted configurations under the stated evaluation procedure. A displayed error identifies a case for closer inspection. A single split cannot establish a general ranking, the best possible score, or reliability on a different population.

B Image measurements

Section 2 supplied two columns so that the tree search could be studied directly. We now reconstruct those columns from the pixels. The corresponding cells of the galaxy classification notebook produce the masks and diagnostic figures.

The extractor measures the scatter of the sky as 1.4826 times the median absolute deviation of the image border, the factor that turns this deviation into a standard deviation for Gaussian noise, plus one grey level. To measure an object, it estimates a background from the image border, keeps the pixels brighter than that background by three times this scatter, removes small structures by a binary opening, and retains the connected component containing the brightest central pixel. A binary opening first strips one layer of pixels from the edge of every bright region and then adds a layer back, so that regions thinner than a few pixels, such as isolated noisy pixels, disappear while larger regions keep their shape. Filling holes gives the mask shown in orange. Measurements use the background-subtracted, nonnegative display brightness inside this mask. Figure B.1 follows observation 5 from pixels to measurements. A mask is a working estimate of the pixels that belong to the object. The masks of observations 4 and 8 include neighbouring sources, and that of observation 5 a faint point source, which distort their measurements. This uncertainty belongs to the input representation before any tree is trained.

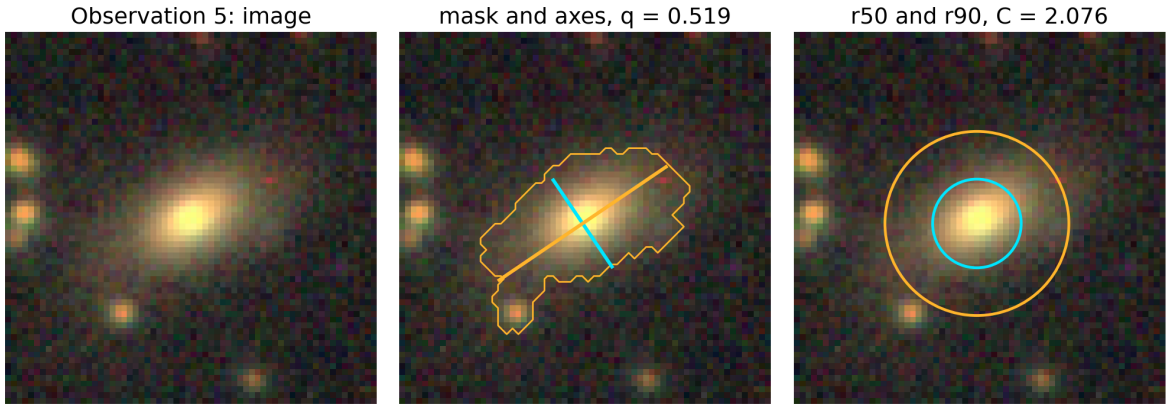


Figure B.1: Observation 5. The image, the mask with the two principal moment axes, and circles enclosing 50% (cyan) and 90% (orange) of the masked display brightness. The axis ratio is $q = 0.519$ and concentration $C = 2.076$. The mask and the numerical features are computed without the class label.

For a precise definition, let $\boldsymbol{r}_p$ be the two-dimensional position of pixel p and w_p its nonnegative, background-subtracted brightness within the mask (zero outside). Define the weighted centre and spatial covariance

$$W = \sum_p w_p, \qquad \bar{\boldsymbol{r}} = W^{-1} \sum_p w_p \boldsymbol{r}_p, \qquad \Sigma = W^{-1} \sum_p w_p (\boldsymbol{r}_p - \bar{\boldsymbol{r}})(\boldsymbol{r}_p - \bar{\boldsymbol{r}})^\top. \tag{B.1}$$

The square roots of the eigenvalues of Σ measure the brightness spread along its principal directions. Thus, for $\lambda_{\min} \le \lambda_{\max}$,

$$q = \sqrt{\lambda_{\min}/\lambda_{\max}}, \qquad C = \frac{r_{90}}{\max(r_{50}, 0.5 \text{ pixel})}. \tag{B.2}$$

Here r_{50} and r_{90} are the radii of the circles about $\bar{\boldsymbol{r}}$ that enclose half and 90% of the total weight W, and the half-pixel floor avoids division by a very small radius. Both features are dimensionless. They describe a masked display composite and carry no calibrated flux or physical size. A tree needs only the resulting columns. Feature extraction and the learning of threshold questions are separate operations.

C Global search and recursive growth

The ideal tree would be found by a global search. For a fixed number L of leaves, among all ways of dividing the measurement space into L boxes $R_1, \ldots, R_L$ we would choose the division that makes the leaves as pure as possible on the fitting data,

$$\min_{R_1,\ldots,R_L} \sum_{\ell=1}^{L} n_\ell \, I(R_\ell), \tag{C.1}$$

where n_ℓ is the number of fitting observations in box R_ℓ and $I(R_\ell)$ their impurity, for regression the mean squared deviation from the leaf mean, so that $n_\ell I(R_\ell)$ is the sum of squared errors of the leaf. Without the limit on L the minimum would be a leaf for every observation, so the number of leaves, or a penalty on complexity, is part of the problem. The number of possible divisions grows so fast with the number of observations and features that this search is out of reach even for modest data sets. Trees are therefore grown by recursive binary splitting [2]. Starting at the root, the algorithm chooses the single best question for the current node, divides the node, and repeats the same choice inside each child, without ever revisiting a question once it is made. The procedure is greedy, since it takes the best step now and gives up the guarantee of reaching the minimum of (C.1). It also restricts the possible divisions to those that a sequence of nested questions can produce, and some box partitions, such as the pinwheel of Figure 8.3 of [2], cannot arise in this way.

D Misclassification error and entropy

Section 4 grew the galaxy tree with the Gini impurity and mentioned two other measures of mixing. The misclassification error is a poor criterion for growing a tree, and the entropy is related to the log loss, as the following derivations show.

A tree that predicts the majority class of each leaf makes a fraction of errors equal to the misclassification error of the leaf,

$$I_M(t) = 1 - \max_k p_{tk}. \tag{D.1}$$

If every observation of the leaf is assigned the same class c, the fraction of errors is $1 - p_{tc}$, which is smallest for the most frequent class. The majority rule (3.2) therefore minimises the error count of a leaf when all mistakes have equal cost. It would seem natural to grow the tree with the same quantity, choosing the question that most reduces the number of errors. Our classifiers are grown with Gini impurity because the error criterion is blind to many useful questions. Entropy is another possible choice.

Suppose that both children of a split have the same majority class k^*. The parent proportions are the mixture $p_{tk} = \pi_L p_{Lk} + \pi_R p_{Rk}$. Since $p_{Lk^*} \geq p_{Lk}$ and $p_{Rk^*} \geq p_{Rk}$ for every k, the mixture also has $p_{tk^*} \geq p_{tk}$, so k^* is the majority of the parent as well. The weighted child error is then

$$\pi_L I_M(L) + \pi_R I_M(R) = 1 - (\pi_L p_{Lk^*} + \pi_R p_{Rk^*}) = 1 - p_{tk^*} = I_M(t), \tag{D.2}$$

and the decrease in misclassification error is zero, however different the children are. The error (D.1) is linear in the proportions as long as the majority class does not change, and a weighted average of a linear function equals the function of the weighted average. The Gini impurity is a strictly concave function of the proportions. For such a function the value at the

mixture $\boldsymbol{p}_t = \pi_L \boldsymbol{p}_L + \pi_R \boldsymbol{p}_R$ exceeds the weighted average $\pi_L I_G(L) + \pi_R I_G(R)$ of the values at the two components unless they coincide (Jensen's inequality), so the Gini decrease is strictly positive whenever $\boldsymbol{p}_L \neq \boldsymbol{p}_R$. Equation (4.14) gives the size of the difference.

A small example shows what is lost. A node holds eight observations of class A and two of class B, so $I_M = 2/10 = 0.2$ and $I_G = 2 \cdot \frac{8}{10} \cdot \frac{2}{10} = 0.32$. A question sends five observations of class A to the left and the remaining three of class A with both of class B to the right. Both children predict A, the two B observations remain misclassified, and the error decrease is zero. The Gini decrease follows from (4.14) with $\pi_L = \pi_R = 1/2$, $\boldsymbol{p}_L = (1, 0)$ and $\boldsymbol{p}_R = (3/5, 2/5)$,

$$\Delta I_G = \frac{1}{2} \cdot \frac{1}{2} \left[\left(1 - \tfrac{3}{5}\right)^2 + \left(0 - \tfrac{2}{5}\right)^2 \right] = \frac{1}{4} \cdot 0.32 = 0.08. \tag{D.3}$$

The split has set aside a pure group and concentrated both B observations in a smaller group, where a later question can separate them. The Gini criterion rewards this preparation and the error criterion does not. The rounder branch of the galaxy tree already shows this mechanism. The supplementary binary diabetes task gives another example of probability predictions and class decisions.

A second concave measure of mixing comes from information theory [9]. The entropy of a node,

$$I_E(t) = -\sum_{k=1}^{K} p_{tk} \log_2 p_{tk}, \tag{D.4}$$

with $0 \log_2 0 = 0$, is the average number of bits needed to communicate the label of an observation drawn from the node. It vanishes for a pure node and reaches its largest value $\log_2 K$ for equal proportions. The entropy decrease (4.4) with $I = I_E$ is called the information gain, and it equals the mutual information between the label and the answer to the question, computed with the empirical proportions of the node. Just as the Gini impurity is the smallest mean squared error of a probability prediction (4.12), the entropy is the smallest mean log loss of a probability prediction $\boldsymbol{v}$. The mean log loss over the node is $-\frac{1}{n_t} \sum_i \ln v_{y_i} = -\sum_k p_{tk} \ln v_k$, and it exceeds its value at $\boldsymbol{v} = \boldsymbol{p}_t$ by $\sum_k p_{tk} \ln(p_{tk}/v_k)$, the Kullback–Leibler divergence, which is nonnegative and vanishes only for $\boldsymbol{v} = \boldsymbol{p}_t$. The minimum is the entropy in natural units, and in bits once ln is replaced by $\log_2$. Figure D.1 compares the three measures for two classes.

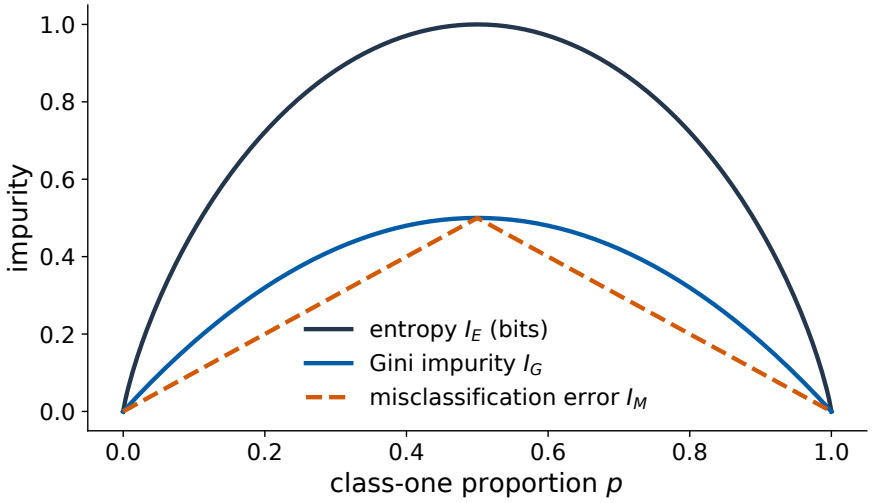


Figure D.1: Entropy in bits, Gini impurity and misclassification error for two classes as functions of the proportion of class 1. The error is piecewise linear with a kink at 1/2. Gini impurity and entropy are strictly concave.

Gini and entropy both vanish at pure nodes and attain their maxima at equal proportions, but they need not rank candidate splits identically. They correspond to different losses, the squared error of probability vectors for Gini and the log loss for entropy.

The difference is largest near purity. For a small minority proportion ε of two classes, the Gini impurity behaves as 2ε and the entropy in natural units as $\varepsilon(1 - \ln \varepsilon)$, which falls more slowly as $\varepsilon \to 0$. Entropy therefore rewards removing the last few minority observations from a node more strongly than Gini does. Gini avoids logarithms and is the default of scikit-learn, and entropy is the criterion of Quinlan's ID3 algorithm [10]. Our single-tree classifiers use Gini and our regression trees use squared error.

E Further datasets

Iris contains four flower measurements and three species labels [11]. Its small table makes the classifier interface easy to inspect. The breast-cancer dataset contains 30 measurements of cell nuclei and a malignant or benign label [12, 13]. Its confusion matrix separates missed malignant cases from false alarms. These datasets provide additional applications of the same fitting and evaluation workflow.

The binary diabetes example labels progression above the median of the fitting responses. Validation and test responses play no part in computing that reference. The label creates a second learning task from the same measurements and carries no diagnostic meaning. Its notebook uses the same short classifier workflow. Detailed pruning and large parameter comparisons are outside the scope of these applications.

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